# FLOATING SOLAR POWER PROJECT BY AMP ENERGY IN MADHYA PRADESH

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|---|---|
| Project title | Floating Solar Power Project by AMP Energy In Madhya Pradesh |
| Project ID | 4785 |
| Crediting period | 06-June-2024 to 05-June-2031 |
| Original date of issue | 12-April-2024 |
| Most recent date of issue | 25-June-2025 |
| Version | 06 |
| *VCS Standard* Version | VCS Standard - v4.7 |
| Prepared by | Ankit Sethiya<br>Manager – Operations<br>EKI Energy Services Limited |

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# 1 PROJECT DETAILS

## 1.1 Summary Description of the Project

AMP Energy Green Seven Private Limited, a wholly owned subsidiary of M/s AMP Solar India Private Limited (the Promoter) is a special purpose vehicle incorporated by AMP Energy Green Private Limited to develop a 140 MWp (100 MWAC) floating solar PV plant (the Project) at Omkareshwar water reservoir located near Punasa Tehsil, Khandwa District, Madhya Pradesh State, INDIA. The purpose of this Project is to utilize the area of the reservoir to generate clean energy & supply the same to M.P. Power Management Company Limited (MPPMCL). A long term PPA (Power Purchase Agreement) had been executed between MPPMCL (Procurer) & AMP (Developer) for 25 years.

The project boundary includes the project power plant (project site where the solar power plant has been installed including the solar power plant, power evacuation infrastructure, energy metering points, switch yards and other civil constructions) implemented in the state and all other power plants/units connected physically to the national (Indian) grid that the project power plants is connected to Unified Indian Grid.

The solar project is designed to achieve substantial greenhouse gas (GHG) emission reductions by utilizing solar energy to generate electricity. By replacing electricity that would otherwise come from fossil fuel-based power plants—such as those fueled by coal, oil, or natural gas—the project significantly reduces GHG emissions. Fossil fuel power plants are known for emitting large quantities of carbon dioxide ($CO_2$) and other GHGs during electricity production.

**A brief description of the scenario existing prior to the implementation of the project**
The baseline scenario would be the continued use of fossil fuel dominated grid connected electricity generation.

This project is commissioned on 06-June-2024 & the expected annual average generation of the project is 211,556 MWh. The emission reduction from the project is estimated to be around 197,846 $tCO_2e$/annum and a cumulative emission reduction of 1,384,927 $tCO_2e$ for the entire crediting period of 7 years.

## 1.2 Audit History

| Audit type | Period | Program | Validation/verification body name | Number of years |
|---|---|---|---|---|
| *Validation* | 06-June-2024 to 05-June-2031 | VCS | LGAI Technological Center, S.A. | Seven Years |

## 1.3 Sectoral Scope and Project Type

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|---|---|
| Sectoral scope[1] | 1 - Energy (renewable/non-renewable) |
| Project activity type | Renewable Energy Project |

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|---|---|
| Sectoral scope | Not applicable as this not a AFOLU project |
| AFOLU project category[2] | Not applicable as this not a AFOLU project |
| Project activity type | Not applicable as this not a AFOLU project |

## 1.4 Project Eligibility

### 1.4.1 General eligibility

As per section 2.1.3 of VCS Standard ver-4.7, table no. 1 states excluded project activities. The exclusion does not apply to concentrated solar thermal-to-electricity, floating solar PV, or energy storage systems e.g., batteries.

This project activity is floating solar PV; Thus, the proposed project is eligible under the scope of the VCS program and the project has been designed to include a single location or installation only.

Project activities supported by a methodology approved under a VCS approved GHG program unless explicitly excluded under the terms of Verra approval: The methodology ACM0002 (version 22.0)[3] is approved under CDM Program, which is a VCS approved GHG program.

It consists in grid-connected electricity generation using (large scale) solar power plant in India, Floating solar is not in Excluded category for Non LDC. In line with the Section 2.1.1 of VCS Standard V4.7.

The project has been designed to include a single location or installation only; it is not being developed as a grouped project. Hence no instance will be added to this project activity in the future.

### 1.4.2 AFOLU project eligibility

Not applicable as this is not a AFOLU project activity.

### 1.4.3 Transfer project eligibility

[1] Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes

[2] See Appendix 1 of the VCS Standard

[3] https://cdm.unfccc.int/UserManagement/FileStorage/R0IJ1X9LQ7W2GOYHSMBFCPE3VKZ685

Not applicable as this project is not transferred or CPA.

## 1.5 Project Design

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

### 1.5.1 Grouped project design

Not Applicable as this is not a grouped project activity

## 1.6 Project Proponent

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|---|---|
| Organization name | AMP Energy Markets India Private Limited |
| Contact person | Abhishek Pandey |
| Title | Director |
| Address | 309, 3rd Floor, Rectangle One, Behind Sheraton Hotel, Saket, New Delhi-110017 |
| Telephone | +91-9717437676 |
| Email | apandey@ampenergyindia.com |

## 1.7 Other Entities Involved in the Project

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|---|---|
| Organization name | EKI Energy Services Limited |
| Role in the project | Project Consultant |
| Contact person | Ankit Sethiya |
| Title | Manager – Operations |
| Address | EnKing Embassy, Plot 48, Scheme 78 Part 2, Vijay Nagar Near Brilliant Convention Centre, Indore–452010, Madhya Pradesh, India |
| Telephone | 0731 428 9086 |

| | |
|---|---|
| Email | ankit@enkingint.org, registry@enkingint.org |

## 1.8 Ownership

The commissioning certificate for project activity is the supporting document to demonstrate that the project ownership lies with M/s AMP Energy Green Seven Private Limited. Although Project owner i.e., M/s AMP Energy Green Seven Private Limited Authorized M/s AMP Energy Markets India Private Limited to act as project proponent for the project activity. This demonstrates the right of use according to clause 3.7.1 (5) of VCS Standard (v4.7) – "An enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals which vests project ownership in the project proponent.

## 1.9 Project Start Date

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| Project start date | 06-June-2024 |
| Justification | The start date is also the date of commissioning of the solar PV project activity |

## 1.10 Project Crediting Period

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|---|---|
| Crediting period | ☒ *Seven years, twice renewable*<br>☐ *Ten years, fixed*<br>☐ *Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)* |
| Start and end date of first or fixed crediting period | 06-June-2024 and 05-June-2031 |

## 1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 $tCO_2e$/year (project)

☐ ≥ 300,000 $tCO_2e$/year (large project)

| Calendar year of crediting period | Estimated GHG emission reductions or removals ($tCO_2e$) |
|---|---|
| 06-Jun-2024 to 31-Dec-2024 | 114,459 |
| 01-Jan-2025 to 31-Dec-2025 | 199,292 |
| 01-Jan-2026 to 31-Dec-2026 | 198,694 |
| 01-Jan-2027 to 31-Dec-2027 | 198,097 |
| 01-Jan-2028 to 31-Dec-2028 | 197,503 |
| 01-Jan-2029 to 31-Dec-2029 | 196,910 |
| 01-Jan-2030 to 31-Dec-2030 | 196,319 |
| 01-Jan-2031 to 05-June-2031 | 83,653 |
| **Total estimated ERRs during the first or fixed crediting period** | **1,384,927** |
| **Total number of years** | **7** |
| **Average annual ERRs** | **197,846** |

## 1.12 Description of the Project Activity

The proposed 140 MWp (100 $MW_{ac}$) floating solar PV plant is developed on the surface of Omkareshwar water reservoir located near Punasa Tehsil, Khandwa District, Madhya Pradesh State, INDIA. Approximately 589 acres of total water cover is available for the proposed 140 MWp (100 $MW_{ac}$) floating solar PV plant using ACM0002 ver-22.0.

The power generated by the floating solar PV plant have evacuated to the utility substation at 220 kV substation Saktapur. Depending on the quantum of power injected into the substation as recorded by the electricity meters at supply feeding point, the state utility i.e., MPPMCL would release the Monthly Generation Reports to the project owner.

The proposed project activity is the installation of a new 100 $MW_{ac}$ floating solar power plant/unit at the site where no renewable power plant was operating prior to the implementation of the project activity (green-field plant)[4].

[4] As per ACM0002 ver-22.0, a Greenfield power plant is defined as a new renewable energy power plant that is constructed and operated at a site where no renewable energy power plant was operated prior to the implementation of the project activity.

Details of proposed solar power project is outlined in the table below:

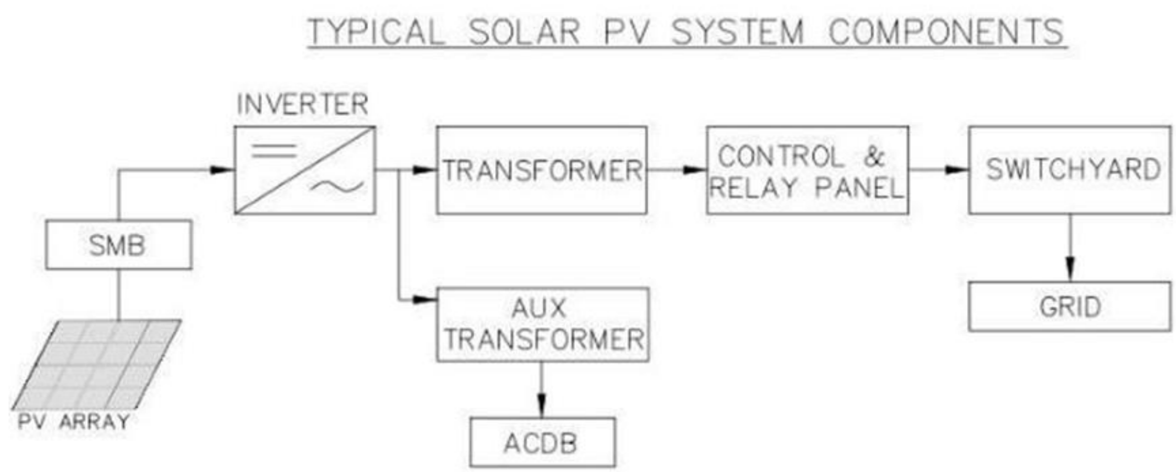
| | |
|---|---|
| Parent Company | M/s AMP Energy Green Private Limited |
| Special Purpose Vehicle | M/s AMP Solar India Private Limited |
| Legal Owner | AMP Energy Green Seven Private Limited |
| Project Proponent | M/s Amp Energy Markets India Private Limited |
| Capacity of the Power Plant | 140 MWp (100 $MW_{ac}$) |
| Date of Commissioning | 06-June-2024 |
| State | Madhya Pradesh |
| Connectivity | Indian National Grid |

The electricity generated from project activity (operational solar power plant) is connected to the National grid. The project activity thus reduces the anthropogenic emissions of greenhouse gases (GHGs) associated with equivalent amount of electricity generation from the existing grid connected power plants (mostly fossil fuel based) and from addition of new generation sources into the grid. The average annual generation of the project is 211,556 MWh. The emission reduction from the project is estimated to be around 197,846 $tCO_2e$ /annum and a cumulative emission reduction of 1,384,927 $tCO_2e$ for the entire crediting period of 7 years.
The operational life of the installed solar module is 25 years based upon the manufacturer's specifications and standard operational & maintenance practices followed at the project site.

Grid connected solar power plant comprises of the main equipment and components listed below.

- Solar PV Modules
- Central inverters
- Module Mounting system
- Grid connect equipment
- Monitoring system
- SCADA
- Cables & connectors

A simple block diagram, related to the interconnection of various systems for grid connectivity, is shown below for reference.




**Solar PV Module –**

A photovoltaic module is a packaged interconnected assembly of photovoltaic cells, which converts sunlight into energy. For this project, poly crystalline PV technology solar module of 540 Wp has been considered.

| Description | Specification |
|---|---|
| Type of PV Module | Mono C-Si Module |
| Solar PV Technology | Monocrystalline |
| Module Make | Longi |
| Power rating (Wp) | 540 Wp or higher |
| Module Power Rating Tolerance (%) | +Ve |
| System Voltage (DC) | 1500 V |
| Temp. Coefficient of power (%/°C) | - 0.34%/Degree C |
| Number of cells | 259,280 |
| Series fuse rating(A) | 30A |
| Glass | Dual Glass, 2+2 heat strengthened glass |
| Module efficiency % | 21.5% |

**Inverter –**

| | |
|---|---|
| **Input DC** | |
| Max. PV input voltage | 1500 V |
| Min. PV input voltage / Start-up input voltage | 960 v/ 990 V |
| MPP voltage range for minimal power | 960 ， 1300 V |
| No. of independent MPP inputs | 1 |
| No. of DC inputs | 30 |
| Max. PV input current | 6112 A |
| **Output AC** | |
| AC output power | 5750 KVA @ 25 degrees Celsius |
| Max. AC output current | 5030 A |
| Nominal AC voltage | 660 V |
| AC voltage range | 561 -726 V |
| Nominal grid frequency / Grid frequency range | 50 Hz / 45 -55 Hz, 60 Hz / 55 - 65 Hz |
| THD | <3 % |
| DC current injection | <0.5 % |
| Power factor at nominal power / Adjustable power factor | >0.99 /0.8 leading |
| **Efficiency** | |
| Max. efficiency | 99.0 % |
| European efficiency | 98.7 % |
| **Protection** | |
| DC input protection | Load break switch + fuse |
| AC output protection | Circuit breaker |
| Overvoltage protection | DC Type I + II / AC Type II |
| Grid monitoring / Ground fault monitoring | Yes / Yes |
| Insulation monitoring | Yes |
| Overheat protection | Yes |
| **General Data** | |
| Dimensions (W*H*D) | 5020 * 2400 * 1100 mm |
| Weight | 6 T |

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|---|---|
| Isolation method | Transformer less |
| Degree of protection | IP65 |

**Module Mounting Structure (MMS)**

HDPE based floats equipped with module mounting arrangements through clamps, nuts and bolts are manufactured, assembled and floated into the water body with modules installed at the bank/shore. The entire island of floating solar is then tugged to the identified location of installation through boat, where the provisions of anchoring and mooring is already made beforehand. The design is made based on site level studies of bathymetric study, geotechnical investigation, etc.

**Monitoring System**

System proposed will maintain and provide all technical information on daily solar radiation availability, hours of sunshine, duration of plant operation and the quantum of power fed to the grid. This will help in estimation of generation and number of other generation parameter of array capacity installed at the site. The system also enables diagnostic and monitoring functions for these components. Communication: Data modem (analogue/Ethernet), few features are presented as follows.

- Monitors the performance of the entire power plant (string wise monitoring, junction boxes, inverters, etc.)
- Evaluates (strings, inverter, nominal/actual value), quantity of DC Power & AC Power produced.
- Measures instantaneous irradiation level and temperature at site. It also measures the module back surface temperature.
- Alerts in case of error (discrepancy in normal operation of components, like module string/ diodes/ inverter/ junction box / box / lose contacts/ etc.) to facilitate recognition and correction of the fault with minimum downtime.
- Visualizes nominal status of the connected components via Control Center PC Software (diagnosis on site or remote)
- Logs system data and error messages for further processing or storing.

**SCADA**

- The instrumentation & control system for the solar power plant will be based on the prevailing standard engineering practices
- Design will ensure full compliances of codes and standards as applicable the field of instrumentation & control for power plant
- The whole plant will be operated through SCADA system

a. The SCADA system shall have the following features:

- Monitoring: Ability to control, using specially designed devices, the state & evolution of one or various physiologic (or others) parameters to detect possible malfunctions
- Remote control: Group of devices which allow modifying the state of the equipment and devices of the plant, from a remote location

b. The SCADA system shall be used for the following minimum tasks:

- To invoice the produced energy
- To detect the incidences and malfunctions (up to logical string level)

- To give the information at a given time interval: a) Availability, b) Performance Ratio and Energy Production

c. The SCADA System will be reliable and robust and some components need redundancy for trouble free operation.

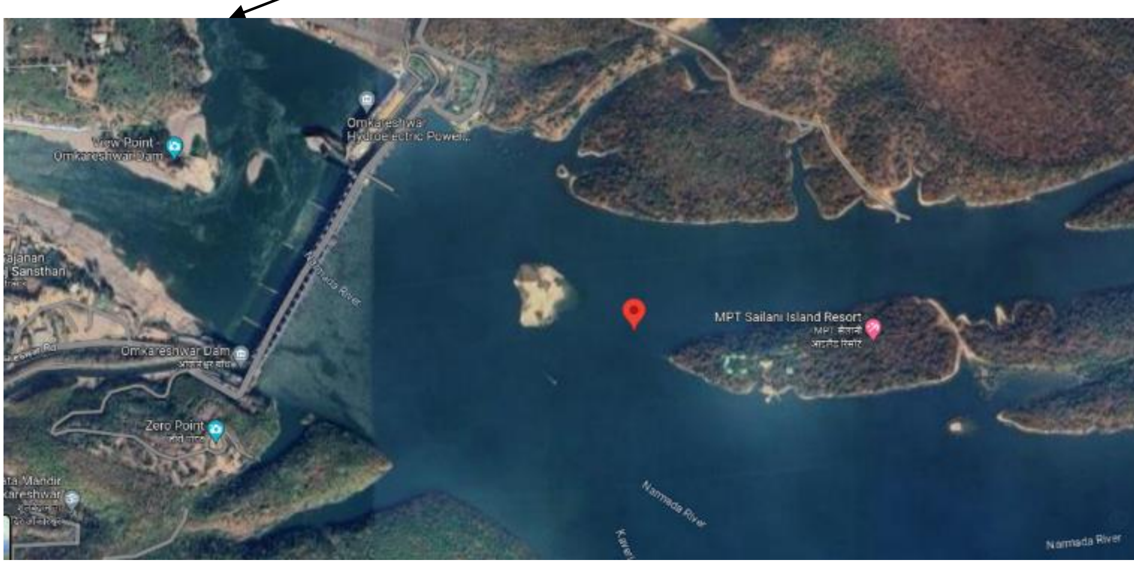
## 1.13 Project Location

The location of the project activity is-

Village – Ghogalgaon
Tehsil – Punasa
District – Khandwa
State – Madhya Pradesh
Country – India

The geographical coordinates are - 22°11'33.0" N (22.1925), 76°13'04.0" E (76.2177)





## 1.14 Conditions Prior to Project Initiation

There was no project at the site of the project proponent prior to the implementation of this project activity. The project proponent was not involved in generation of solar PV based power and supplying it to the grid under the pre-project scenario, therefore, in the absence of the proposed project activity, the equivalent amount of electricity would have been generated from the connected / new power plants in the Unified Indian Grid, while proposed project activity is a green field project. Therein, the main emission source in the pre-project scenario is the power plants connected to the Unified Indian Grid and the primary GHG involved is $CO_2$.

## 1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The Project has received necessary approvals for development and commissioning from the state Nodal agencies and is in compliance to the local laws and regulations.

The relevant national laws and regulations pertaining to generation of energy in India are:

- Electricity Act 2003[5]
- National Electricity Policy 2005[6]

[5] https://cercind.gov.in/Act-with-amendment.pdf

[6] https://indiatransmission.org/listdocuments/Policy/NEP?doc_id=1101

- Tariff Policy 2006[7]
- The Project activity conforms to all the applicable laws and regulations in India:
- Power generation using solar energy is not a legal requirement or a mandatory option.
- There are state and sectoral policies, framed primarily to encourage solar power projects. These policies have also been drafted realizing the extent of risks involved in the projects and to attract private investments.
- The Indian Electricity Act, 2003 (May 2007 Amendment) does not influence the choice of fuel used for power generation.
- There is no legal requirement on the choice of a particular technology for power generation.

**Impact on Terrestrial Ecology & Marine biodiversity**

The solar modules will be installed upon the water surface shall have close proximity to each other and therefore can mirror the reflection of a large water body during daytime hours. The risk of the lake effect wherein birds and bats mistake the solar modules as a water body is more pronounced in this Solar Park. After development of floating platforms, there will be distinct separation from water body to platforms, thus disturbance issue would be reduced due to the water birds.

The reservoir ecology has less amount of biodiversity and richness of species and the water system is not a natural flowing system. The organic content of the reservoir sediments also found low as the system can be considered as new and developed artificially after development of the Dam. Majority of the fish species are commercial and artificially cultivated for commercial fishing. Installation of solar panel only in limnetic part of the reservoir, where under water biodiversity is considered as low. Maintain minimal activity in Littoral zone.

The floating solar park will be properly fenced with flatting fencing facility so that floating solar park can be separated properly from the free part of the reservoir, where fishermen can get access for fishing. Further, all the pooling stations will be protected with 24 hours of security facility.

As the area does not fall under any flyway for migratory birds, the assemblage of aquatic and wetland birds was truncated. The richness of such birds in terms of abundance and diversity was low and patchily distributed. The birds were only encountered near littoral zones, as a usual behaviour in larger water bodies. The nutrient-rich littoral zone and shoreline provide essential food and shelter for the foraging birds, therefore, the activities of the aquatic birds, such as foraging, roosting, resting, were restricted to the littoral zone of the reservoir.

fish movement would not be altered due to the development of floating solar PV park on the reservoir and majority fish population shall be available along the littoral zones of the reservoir. Thus, fish catch and commercial fishing activity and fishermen’s boat movement shall be restricted towards littoral zone only but impact on fish catch can be considered as moderate.

## 1.16 Double Counting and Participation under Other GHG Programs

### 1.16.1 No Double Issuance

[7] https://cea.nic.in/wp-content/uploads/legal_affairs/2020/09/Tariff%20policy.pdf

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

### 1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

### 1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

## 1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

### 1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

### 1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

### 1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

## 1.18 Sustainable Development Contributions

The project involves installation of Solar Photovoltaic (SPV) Panels on the surface of Omkareshwar water reservoir located near Punasa, Khandwa District, Madhya Pradesh State, INDIA. The project thus replaces fossil fuel dominated grid electricity thereby reducing the equivalent amount of GHG emission and contributing to sustainable development. The project activity has also provided employment to local people, thus contributing towards SDG 8. The sustainable development contributions made by the project activity are mentioned below.

**Contribution to sustainable development:**

Apart from generation of renewable electricity, the project activity would contribute to the sustainable development of the region - socially, environmentally and economically. Ministry of Environment, Forest and Climate Change, has stipulated economic, social, environment and technological well-being as the four indicators of sustainable development. The project contributes to sustainable development through the following ways.

- **Social well-being:** The project would help in generating employment opportunities during the construction and operation phases. The project activity will lead to utilization of unused area with which the sources of development increases. This project will lead to development in infrastructure in the region by the multiple utilizations of single source and also may promote business with improved electricity generation.
- **Economic well-being:** The project is a clean technology investment in the region, which would not have been taken place in the absence of the VCS benefits & the project activity will also help to reduce the demand supply gap in the state. The project activity creates local employment generation which helps economic well-being of local people.
- **Technological well-being:** The successful operation of project activity would lead to promotion of floating solar based power generation and would encourage other entrepreneurs to participate in similar projects.
- **Environmental well-being:** Solar being a renewable source of energy, it reduces the dependence on fossil fuels and conserves natural resources which are on the verge of depletion. Due to its zero emission the Project activity also helps in avoiding significant amount of GHG emissions.

The sustainable development contributions made by the project activity are mentioned below –

The total estimated emission reductions during the entire crediting period of 7 years are approximately 1,384,927 $tCO_2e$.

- For SDG -7. The project activity is in operation from the date of commissioning & this SDG will be accounted during the verification. Estimated average annual generation from the project activity is 211,556 MWh.
- For SDG -8. The project activity has created 22 employments and 1training during year shall be imparted to the employees.
- For SDG -13. The project activity is in operation from the date of commissioning & this SDG will be accounted during the verification. Estimated average annual emission reductions from the project activity is 197,846 $tCO_2e$.

## 1.19 Additional Information Relevant to the Project

### 1.19.1 Leakage Management

Since the project activity is a renewable energy project based on solar technology, no leakage emissions are considered. Thus, leakage management plan and implementation of leakage and risk mitigation measures are not applicable for this project activity.

### 1.19.2 Commercially Sensitive Information

Not applicable. No any commercially sensitive information has been excluded from the public version of the project description.

### 1.19.3 Further Information

Not Applicable

# 2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

## 2.1 Stakeholder Engagement and Consultation

### 2.1.1 Stakeholder Identification

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| Stakeholder Identification | Following stakeholders have been identified for this project activity by analyzing the beneficiaries /people who are directly affected in daily life by the project activity, impact of local/regional /provincial/national government policies in implementation and operation of the project activity, other groups having social/political influences in the project location<br>1. Local community members<br>2. Local government officials<br>3. Local media personnel<br>4. Regional and national government officials<br>5. Local NGOs.<br>6. Local tribal resident near the project area which is Bilaya also a part of stakeholder consultation, Bilaya had a mixed population, |

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|---|---|
| | where the Scheduled Tribes were primarily engaged in fishing. The village has also been resettled after displacement by the construction of Omkareshwar Dam. |
| Legal or customary tenure/access rights | Employees have access to the project site and access to the resources (as per their designation/work profile). Local community members have access to project site if they want to share any VCS Joint Project feedback/grievance about the project activity (implementation and operation)/ Local NGOs/representatives from media are welcome to the project with prior appointment. Local/provincial;/national government officials can also visit the project for any inspection regarding compliance to the regulations or any other purpose as required by the government.<br><br>The project is located at the water surface of dam and the legal or customary tenure or rights access with the NHDC (Narmada Hydro development corporation) joint venture of Madhya Pradesh state government. The area is prohibited for the entry of unauthorized people that may be local communities and indigenous people due to safety precautions. |
| Stakeholder diversity and changes over time | Stakeholders identified for this project activity are diverse in nature – in terms of socio-economic status, age and gender wise and the diversity is unlikely to change over time. |
| Expected changes in well-being | Expected changes<br><br>1. Economic well-being – the project activity is likely to provide employments to the locality thus economic well beings are expected<br><br>2. Environmental well-being – as this project activity cuts down GHG emissions in the atmosphere generating electricity using solar energy based solar PV technology; thus, it will bring environmental well-being in the locality |
| Location of stakeholders | There is no impact by the project activity to the location of stakeholders, Indigenous Peoples (IPs), local communities (LCs), customary rights holders, and areas outside the project area. |

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| Location of resources | Not Applicable as this project activity does not contain any location of territories and resources which stakeholders own or to which they have customary access. |

### 2.1.2 Stakeholder Consultation and Ongoing Communication

<table>
<tr><td>Date of stakeholder consultation</td><td>05-December-2023</td></tr>
<tr><td>Stakeholder engagement process</td><td>Identified stakeholders were local community members i.e., local villagers, employees of the company, local government officials (village members), local NGO workers and media personnel.<br><br>Stakeholders were invited through public notice and telephone communications – local community members were invited through public notice while local government, local NGO workers and media personnel were invited mostly by telephonic communications.<br><br>Date of Invitation: 20-November-2023</td></tr>
<tr><td>Consultation outcome</td><td>The meeting location – Kelwa Buzurg Village, Punasa Tehsil, Madhya Pradesh State<br><br>The purpose of the meeting was clearly outlined i.e., making local stakeholders including employees of the company, local administration representatives, local community, NGOs working for environmental cause etc., gets the awareness of this solar project activity that involves electricity generation using solar energy through solar PV technology. The primary objective of this project is generating the clean energy by reducing emissions of GHGs in comparison to pre-project or baseline scenario.<br><br>The PP was encouraged by all the positive feedback/ comments received from stakeholders and there were no negative comments in any manner. The questions asked by stakeholders were only on process of electricity generation, which was clearly elaborated by PP representatives.<br><br>Opportunities given to stakeholder-<br><table><tr><td>Comments</td><td>Response</td></tr><tr><td>The people of the village demanded livelihood opportunity during project</td><td>The consultant team assured them that unskilled employment shall be provided</td></tr></table></td></tr>
</table>

| | | |
|---|---|---|
| | construction and operation phase. | to local people based on their skill and education during project construction phase. But during project operation phase, there is minimal employment opportunity. |
| | The community requested for basic infrastructure in the village like electricity and road | The ESIA team said that the electrification and construction of road is not under development of solar scheme but they shall raise the issue at right platform. |
| | The people of the village enquired whether there is any land acquisition proposed and route of Transmission line. | The Consultant team ensured that there is no any land acquisition involved in the project. The proposed project is planned on Omkareshwar reservoir. The land required for pooling sub-station are proposed on government land. The ESIA team explained proposed tentative route of transmission line. |
| | People concern about their present dairy business in town | Small investment for cattle rearing and linkages to dairy aggregators may be explored |
| | Restoring Livelihood through providing alternative livelihood opportunities for women, Providing equal opportunities in the existing work at preconstruction and construction and if possible, Operation phases as well | The suggested activity in terms of compliance shall be included in the ESMP to ensure gender equality and better and safe participation of women in income generating activities during preconstruction and construction phase.<br><br>The suggested activity in terms of compliance shall be included in the ESMP to ensure gender equality and better and safe participation of women in income generating activities during pre-construction and construction phase. |

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|---|---|
| Ongoing communication | Stakeholders' consultation is a continuous process. To attract or encourage the stakeholders to raise their issues or suggestions to the management at any time, a feedback register is maintained at the project site office. It is accessible to all the stakeholders & they simply can walkdown to record their suggestion/feedback or complaint. Irrespective of the nature of the complaint, the same will be addressed with utmost priority. |
| Stakeholder input | The feedbacks/comments received from stakeholders were quite positive and no negative comments were received. Hence, it was not necessary for any updates to project design as the questions asked by stakeholders were only on process of electricity generation, employment opportunities to the local people and these were explained satisfactorily to stakeholders by the PP representatives. |

### 2.1.3 Free Prior and Informed Consent

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|---|---|
| Obtaining consent | Not Applicable as the project is located in PP's property, so it is not affecting IPs, LCs, and customary rights holders. Hence consent not required. |
| Outcome of FPIC | Not Applicable, as no FPIC is required. |

### 2.1.4 Grievance Redress Procedure

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|---|---|
| Development process | Grievance/Feedback register is available at project site office where any stakeholder can come and register his/her feedback along with their particulars. |
| Grievance redress procedure | Stakeholders' feedback is monitored daily and any grievance recorded is resolved immediately & the same is informed to the respective stakeholder. |

### 2.1.5 Public Comments

This project was open for public comments[8] from 13-February-2024 to 14-March-2024 on Verra Registry.

| Comments received | Actions taken |
|---|---|

[8] https://registry.verra.org/app/projectDetail/VCS/4785

| | |
|---|---|
| There were no comments received from the stakeholders regarding project activity. | Not applicable as there are no comments till date i.e., 08-July-2024 (outside the public commenting period also) |

## 2.2 Risks to Stakeholders and the Environment

| | Risks identified | Mitigation or preventative measure taken |
|---|---|---|
| Risks to stakeholder participation | No risk to stakeholder participation is anticipated | There are no exclusions, lack of communications or barriers in the project activity that might prevent stakeholders from engaging or participating. Hence, no mitigation or preventive measure is required as no risk has been identified |
| Working conditions | No risk identified | PP has taken all measures to ensure proper working condition by implementing EHS policy[9] |
| Safety of women and girls | No risk identified | PP ensures safety of women and girls in the project site and abide by the relevant national laws |
| Safety of minority and marginalized groups, including children | No risk identified | PP ensures the protection of minority and marginalized communities, including children[10], at the project site in compliance with local regulations. |
| Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials) | No risk identified | There is negligible pollution in air, noise, water due to project activity implementation and operation. |

## 2.3 Respect for Human Rights and Equity

9 https://labour.gov.in/sites/default/files/doc1.pdf
10 https://labour.gov.in/sites/default/files/act_2.pdf

### 2.3.1 Labor and Work

| | |
|---|---|
| Discrimination and sexual harassment | PP confirmed that there is no discrimination or sexual harassment occurs at workplace & PP created a workplace that is free of sexual harassment and discrimination following the act "THE SEXUAL HARASSMENT OF WOMEN AT WORKPLACE (PREVENTION, PROHIBITION AND REDRESSAL) ACT, 2013[11]" stipulated by host country (India) government and same has been demonstrated during validation assessment process. |
| Management experience | The management team of the project has the experience and expertise to effectively implement similar project activities. The team members have many years of working experience in the field of Solar power installation and environmental protection, and have relevant skills and expertise in project planning, implementation and supervision to effectively manage and drive the implementation of the project. Manager of the project has experience of engaging communities and the management team will be trained for engaging communities.<br><br>In addition, PP has developed a recruitment strategy to fill gaps in skills and experience that may exist in the team and will be looking for people with relevant experience and expertise to join the team to ensure that the project is supported and guided. |
| Gender equity in labor and work | PP confirmed that there is no discrimination in workers with respect to gender in providing jobs and also PP ensures that there is no discrimination in respect to wages between genders by following the "Advisory for Employers to Promote Women Workforce Participation"[12]. This has been demonstrated by the project proponent during validation assessment process. |
| Human trafficking, forced labor, and child labor | PP has confirmed that project activity does not and will not use victims of human trafficking, forced labour, and child labour by following the national policy "THE CHILD LABOUR (PROHIBITION AND REGULATION) ACT, 1986"[13] stipulated by host country (India) government. This has been demonstrated by the project proponent during validation assessment process. |

### 2.3.2 Human Rights

[11] https://doe.gov.in/files/inline-documents/DoE_Prevention_sexual_harassment.pdf

[12] https://labour.gov.in/sites/default/files/012524_booklet_ministry_of_labour_employement_revised2.pdf

[13] https://labour.gov.in/sites/default/files/act_2.pdf

PP has provided declaration that there will be no human rights violation of any kind to any of the stakeholders during construction and operation of the project activity. Protection of Human Rights Act, 1993[14]. Long Title: An Act to provide for the constitution of a National Human Rights Commission, State Human Rights Commissions in States and Human Rights Courts for better protection of human rights and for matters connected therewith or incidental.

### 2.3.3 Indigenous Peoples and Cultural Heritage

PP is committed to protect regional as well as national cultural heritage.

### 2.3.4 Property Rights

| | |
|---|---|
| Rights to territories and resources | Not applicable as this is a floating solar project and 100% owned by the project legal owner. Hence, there are no any legal or customary tenure/access rights to territories, property, and resources, including collective and/or conflicting rights, held by stakeholders. |
| Respect for property rights | Not applicable as this is a floating solar project which is 100% owned by the project legal owner & no further measures are required in respect to property rights. |

### 2.3.5 Benefit Sharing

| | |
|---|---|
| Process used to design the benefit sharing plan | Public information and consultations were carried out during the project pre-construction stage in the form of community meeting, focused group discussion, in-depth interviews and individual consultations. The consultation process ensured that the likely project affected persons (PAPs), local community and other stakeholders were informed in advance and allowed to participate actively and consulted. This serves to reduce the insecurity among local community and likely PAPs and thereby opposition to the project because of its transparent nature inbuilt in the consultation process |
| Summary of the benefit sharing plan | A benefit-sharing plan for a floating solar plant typically involves distributing the economic, environmental, and social benefits among various stakeholders. Here's a summary of the key components: |

[14] https://www.indiacode.nic.in/bitstream/123456789/15709/1/A1994____10.pdf

| | |
|---|---|
| | 1. **Economic Benefits**:<br>○ **Revenue Sharing**: Profits from the sale of electricity generated by the floating solar plant can be shared among investors, local communities, and government entities.<br>○ **Job Creation**: Employment opportunities for local communities in the construction, operation, and maintenance of the plant.<br>○ **Reduced Energy Costs**: Lower electricity costs for nearby communities and industries due to the increased supply of renewable energy.<br>2. **Social Benefits**:<br>○ **Community Development**: Investments in local infrastructure, education, and healthcare as part of corporate social responsibility initiatives.<br>○ **Energy Security**: Increased energy independence for the region, reducing reliance on fossil fuels and imported energy.<br>○ **Public Awareness**: Educational programs and awareness campaigns to promote the benefits of renewable energy and environmental conservation. |
| Approval and dissemination of benefit sharing plan | The Floating Solar Park will be developed under the Government of India's Solar Park Scheme, and will be eligible for all of the scheme's financial benefits. MNRE will give a grant under the 'Scheme for Development of Solar Parks and Ultra Mega Solar Power Projects' as part of the park's development. |

## 2.4 Ecosystem Health

The municipal waste from the site office will only be routed through proper collection and handover to local municipal body authorized waste management facility for further disposal. There will be only periodic generation of hazardous wastes/ E-waste during any repair and O&M activity. The waste will be disposed of through approved vendors in accordance to Hazardous Wastes Rules, 2016[15]. The solar panels made up off antimony containing glasses, hence, improper disposal of used/broken solar panel may cause harm to environment. Thus, hierarchy of control should include elimination of hazardous. material containing panels; contractor must ensure the solar panel without the hazardous material are used for the installation to avoid a later problem with disposal.

Hazardous materials and waste will not be stored near any drainage channels or cliff-sides to prevent contamination of the surrounding environment and impact on local flora/fauna.

| | Risks identified | Mitigation or preventative measure taken |
|---|---|---|
| Impacts on biodiversity and ecosystems | No risk identified | It is envisaged that the impact on littoral zone is greater than the limnetic zone. The ecological impact of construction activities will be High in littoral zones and Moderate in limnetic zone. fishing activities are limited to certain specific areas which chiefly occupy the lower-depth areas near littoral zone. They fish yield in the deeper sections of the reservoir is less, which can be translated as – congregation of smaller school of fishes in the limnetic zone.<br>It is hypothetically predicted that the reflection from solar models may disturb the aquatic birds, as they may perceive panels as water body and may accidentally land on them. However, it is also assessed that, firstly, the reflectance of panels will be different from that of water and, secondly, the panels situated in the middle section of the reservoir will naturally be avoided by birds, therefore, the impact will be low. |

[15] https://www.iwma.in/HWM%20Rules.pdf

| | | |
|---|---|---|
| | | No impact on biodiversity is likely to be caused as the location of project activity is not identified as the biodiversity prone area by any national or local government institutes[16]. |
| Soil degradation and soil erosion | No risk identified | No soil erosion and degradation are likely to be caused as this is a floating solar project only. |
| Water consumption and stress | No risk identified | Water consumption will be of moderate level and hence stress is likely to be caused. |
| Usage of fertilizers | No risk identified | As it is a floating solar power project, the usage of fertilizers is zero. |

### 2.4.1 Rare, Threatened, and Endangered species

*Is the project located in or adjacent to habitats for rare, threatened, or endangered species?*

☐ Yes ☒ No

### 2.4.2 Introduction of species

According to the PD filling guidelines of Version 4.3, the table in this section is not required for the projects with no planting or species introduction & this section may be indicated as Not Applicable.

### 2.4.3 Ecosystem conversion

This is not an ARR, ALM, WRC or ACoGS project. Hence, there is no Ecosystem conversion is involved in this project activity.

# 3 APPLICATION OF METHODOLOGY

## 3.1 Title and Reference of Methodology

16 https://www.bbau.ac.in/Docs/FoundationCourse/TM/Biodiversity%20Hotspots%20in%20India.pdf (Source : Conservation International : www.conservation.org ; www.cepf.net)

| Type (methodology, tool or module). | Reference ID, if applicable | Title | Version |
|---|---|---|---|
| Methodology | ACM0002 | Grid-connected electricity generation from renewable sources[17] | 22.0 |
| Tool | 01 | Tool for the demonstration and assessment of additionality[18] | 07.0.0 |
| Tool | 05 | Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation[19] | 03.0 |
| Tool | 07 | Tool to calculate the emission factor for an electricity system[20] | 07.0 |
| Tool | 24 | Common Practice[21] | 03.1 |
| Tool | 27 | Investment analysis[22] | 14.0 |

## 3.2 Applicability of Methodology

The project activity is a green field grid connected floating solar power project with cumulative capacity of 140 MWp (100 $MW_{ac}$). Hence, Consolidated Methodology ACM0002: Grid-connected electricity generation from renewable sources is applied. The applicability of this methodology is established through assessment of the following conditions.

| Methodology ID | Applicability condition | Justification of compliance |
|---|---|---|
| ACM0002, Version 22.0 | Para 05:<br>This methodology is applicable to grid-connected renewable energy power generation project activities that: | The project activity is installation of a greenfield floating solar power (renewable) project.<br><br>Hence the project activity meets the applicability condition of the methodology. |

[17] https://cdm.unfccc.int/UserManagement/FileStorage/R0IJ1X9LQ7W2GOYHSMBFCPE3VKZ685
[18] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf

[19] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf

[20] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf
[21] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf
[22] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v14.0.pdf

| | | |
|---|---|---|
| | a) Install a Greenfield power plant;<br>b) Involve a capacity addition to (an) existing plant(s);<br>c) Involve a retrofit of (an) existing operating plant(s)/unit(s);<br>d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or<br>e) Involve a replacement of (an) existing plant(s)/ unit(s).<br>f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid. | |
| ACM0002, Version 22.0 | Para 07:<br>In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that:<br>a) Integrate BESS with a Greenfield power plant;<br>b) Integrate a BESS together with implementing a capacity addition to (an) existing solar | The project activity does not involve the integration of BESS hence this condition is not applicable. |

| | | |
|---|---|---|
| | photovoltaic1 or wind power plant(s)/unit(s);<br>c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s);<br>d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/ unit(s).<br>*e)* Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant. | |
| ACM0002, Version 22.0 | Para 08:<br>The methodology is applicable under the following conditions:<br>a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/ unit, wave power plant/unit or tidal power plant/unit;<br>b) In the case of capacity additions, retrofits, | The project activity involves construction and operation of greenfield grid-connected floating solar power project. Hence, complies to this applicability condition (a).<br>Since the project activity does not include capacity additions, retrofits, rehabilitations, or replacements of existing plant/unit the applicability condition (b) is not applicable/ relevant for the project activity.<br>The project activity does not involve the integration of BESS hence this condition (c) is not applicable.<br>The project activity does not involve the integration of BESS hence this condition (d) is not applicable. |

| | | |
|---|---|---|
| | rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity;<br>c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents);<br>d) The BESS should be charged with electricity generated from the associated renewable energy | The project activity does not involve the PSP; hence this condition (e) is not applicable. |

| | | |
|---|---|---|
| | power plant(s). Only during exigencies 2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.<br>e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in | |

| | | |
|---|---|---|
| | coordination with the PSP during pumping mode. | |
| ACM0002, Version 22.0 | Para 09:<br>In case of hydro power plants, one of the following conditions shall apply:<br>a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs;<br>b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, is greater than 4 $W/m^2$<br>c) The project activity results in new single or multiple reservoirs and the power density is greater than 4 $W/m^2$<br>d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, is lower than or equal to 4 $W/m^2$ and all of the following conditions shall apply:<br>(i) The power density calculated using the total installed capacity of the integrated project is greater than 4 $W/m^2$; | The project activity involves construction and operation of greenfield grid-connected floating solar power project using solar energy for generation of electricity hence this applicability condition is not applicable/relevant to the project activity as the applicability conditions is related to hydro power projects |

| | | |
|---|---|---|
| | (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity;<br>(iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 $W/m^2$ are:<br>a. Lower than or equal to 15 MW; and<br>b. Less than 10 per cent of the total installed capacity of integrated hydro power project. | |
| ACM0002, Version 22.0 | Para 10:<br>In the case of integrated hydro power projects, project participants shall:<br>a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or<br>b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. | The project activity involves construction and operation of greenfield grid-connected floating solar power project using solar energy for generation of electricity; hence this applicability condition is not applicable/relevant to the project activity as the applicability conditions is related to hydro power projects. |
| ACM0002, Version 22.0 | Para 11: | This project is not a PSP. Hence, this condition is not applicable. |

| | | |
|---|---|---|
| | In the case of PSP, the project participants shall demonstrate in the PDD that the project is not using water which would have been used to generate electricity in the baseline. | |
| ACM0002, Version 22.0 | Para 12:<br>The methodology is not applicable to:<br>a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site;<br>b) Biomass fired power plants/units. | The project activity involves construction and operation of greenfield grid-connected floating solar power project using solar energy for generation of electricity hence this applicability condition is not relevant as the same pertains to switching from fossil fuels to renewable energy sources or biomass fired power plants/units. |
| ACM0002, Version 22.0 | Para 13:<br>In the case of rehabilitations, retrofits, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance" | The project activity involves construction and operation of greenfield grid-connected solar power project using solar energy for generation of electricity hence the applicability condition "10" is not relevant as the same pertains to retrofits, rehabilitations, replacements, or capacity additions. |
| ACM0002, Version 22.0 | Para 14: | The applicability of the tools is outlined below. |

| | | |
|---|---|---|
| | In addition, the applicability conditions included in the tools referred to below apply | |

**TOOL01: Tool for the demonstration and assessment of additionality - Version 7.0.0 (EB 70, Annex 08)**

| Methodology ID | Applicability condition | Justification of compliance |
|---|---|---|
| TOOL01: Version 07.0.0 | Para 09:<br>The use of the "Tool for the demonstration and assessment of additionality" is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool. | This condition is not applicable as the project proponent is not proposing new methodologies. |
| TOOL01: Version 07.0.0 | Para 10:<br>Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory. | This condition is applicable as project activity is applying this additionality tool. |

**TOOL05: Tool for the demonstration and assessment of additionality - Version 7.0.0 (EB 96, Annex 05)**

| Methodology ID | Applicability condition | Justification of compliance |
|---|---|---|
| TOOL05: Version 03.0 | Para 05:<br>If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: | The emissions are not calculated for electricity consumption in this project; hence, these scenarios & this applicability condition is not applicable. |

| | | |
|---|---|---|
| | (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer;<br>(b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or<br>(c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid. | |

| | | |
|---|---|---|
| TOOL05: Version 03.0 | Para 06:<br>This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated:<br>(a) Scenario I: Electricity is supplied to the grid;<br>(b) Scenario II: Electricity is supplied to consumers/ electricity consuming facilities; or<br>(c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities. | This tool is referred in line with the methodology "ACM0002: Grid connected electricity generation from renewable sources", Version 22.0. Scenario I is applicable for the project activity as the electricity generated from the grid is being supplied to the grid. |
| TOOL05: Version 03.0 | Para 07:<br>This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for $CO_2$ emissions. | The solar power project is installed to supply electricity to the national grid and not to provide electricity in the project activity. Hence, this condition is not applicable. |

**TOOL07: Tool to calculate the emission factor for an electricity system - Version 07.0 (EB 100, Annex 04)**

| Methodology ID | Applicability condition | Justification of compliance |
|---|---|---|
| TOOL07: Version 07.0 | Para 03:<br>This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid | The project is a Greenfield solar power project that results in savings of electricity that would have been provided by the grid and thus the tool is applicable. |

| | | |
|---|---|---|
| | electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects). | |
| TOOL07: Version 07.0 | Para 04:<br>Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in "Appendix 2: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and | Steps involved in calculation of Emission Factor is included in section 4.1 of this VCS PD as per the requirement of the tool. |

| | | |
|---|---|---|
| | not to other aspects such as transmission capacity. | |
| TOOL07: Version 07.0 | Para 05:<br>In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country. | Project activity is located in non-Annex I country and hence the tool is applicable. |
| TOOL07: Version 07.0 | Para 06:<br>Under this tool, the value applied to the $CO_2$ emission factor of biofuels is zero. | The individual project activity is a solar project and there is no involvement of biofuels. |

TOOL24: Common Practice – Version 03.1 (EB 84, Annex 07)

| Methodology ID | Applicability condition | Justification of compliance |
|---|---|---|
| TOOL24: Version 03.1 | Para 03:<br>This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality. | The proposed project activity is applying the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality. |
| TOOL24: Version 03.1 | Para 04:<br>In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the | The applied approved baseline and monitoring methodology does not define approaches for the conduction of the common practice test that are different from those described in this methodological tool. |

| | | |
|---|---|---|
| | requirements contained in the methodology shall prevail. | |

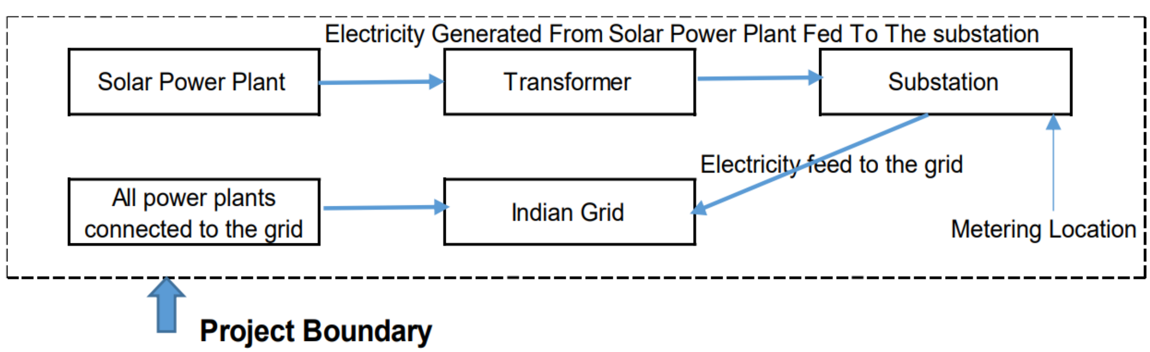
**TOOL27: Investment analysis – Version 14.0 (EB 122, Annex 01)**

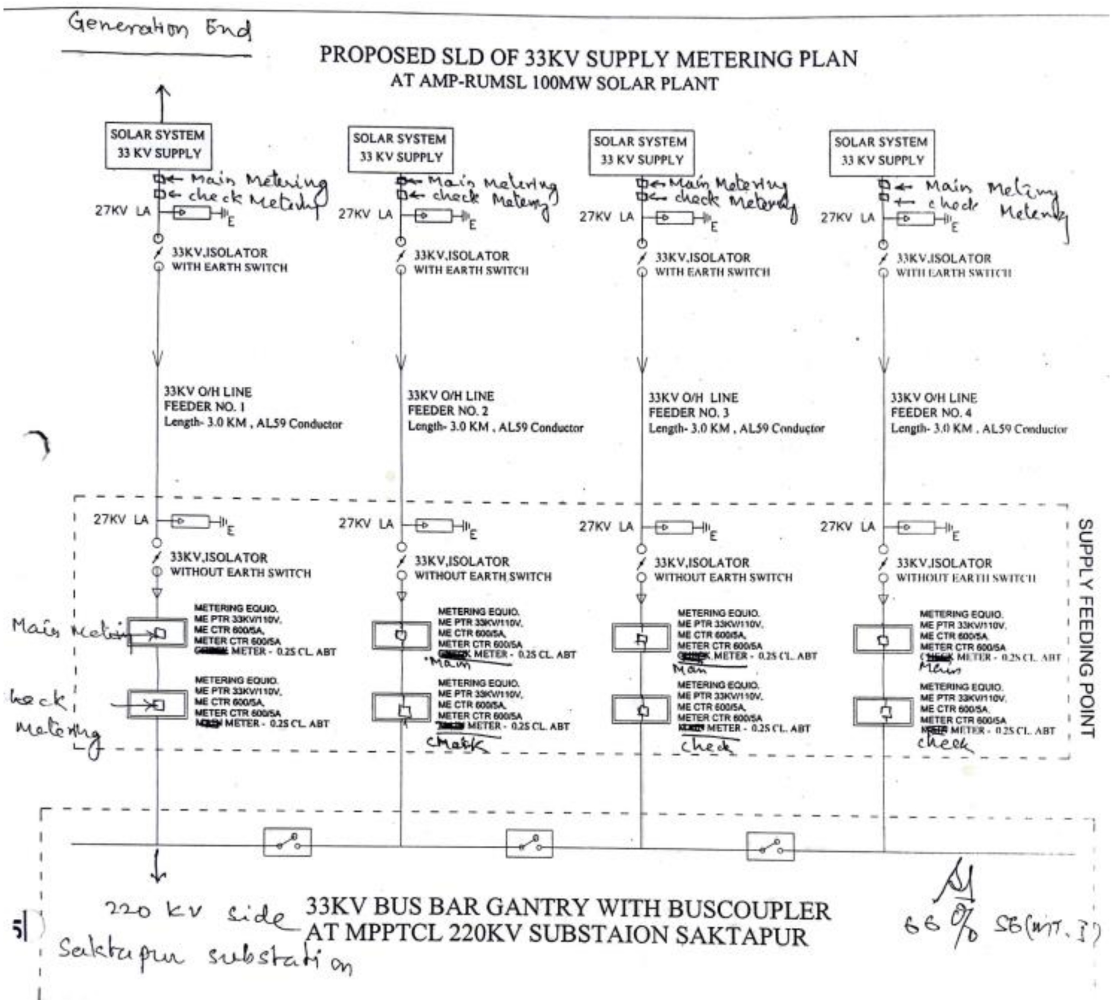
| Methodology ID | Applicability condition | Justification of compliance |
|---|---|---|
| TOOL27: Version 14.0 | Para 02:<br>This methodological tool is applicable to CDM project activities and programmes of Activities (PoAs) that conduct an investment analysis for the demonstration of additionality and/or the identification of the baseline scenario. | This condition is applicable. |
| TOOL27: Version 14.0 | Para 03:<br>In case the applied approved baseline and monitoring methodology contains requirements for investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail. | This condition is not applicable. The applied methodology, ACM0002, version 22.0 does not contain any requirements for the investment analysis that are different from those described in this methodological tool. |

## 3.3 Project Boundary

The spatial extent of the project boundary includes the project power plant / unit and all power plants / units connected physically to the electricity system that the project power plant is connected to. The project boundary therefore includes the project site where the power plant has been installed, associated power evacuation infrastructure, energy metering points, switch yards the national grid of India.

**Single Line diagram of project activity:**






| Source | | Gas | Included? | Justification/Explanation |
|---|---|---|---|---|
| Baseline | Grid connected electricity generation | $CO_2$ | Yes | Main emission source |
| | | $CH_4$ | No | Minor emission source |
| | | $N_2O$ | No | Minor emission source |
| | | Other | No | Project activity does not emit other forms of GHG emissions |
| Project | $CO_2$ emissions from the | $CO_2$ | Yes | Main emission source |
| | | $CH_4$ | No | Project activity does not emit $CH_4$ |

| Source | | Gas | Included? | Justification/Explanation |
|---|---|---|---|---|
| | consumption of grid electricity used as auxiliary power to the project activity | $N_2O$ | No | Project activity does not emit $N_2O$ |
| | | Other | No | Project activity does not emit other forms of GHG emissions |

## 3.4 Baseline Scenario

As per the approved consolidated methodology ACM0002, Version 22.0, para 27, If the project activity is the installation of a Greenfield power plant with or without a BESS, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "TOOL07: Tool to calculate the emission factor for an electricity system".

The project activity involves setting up of solar projects to harness the solar energy to produce electricity and supply to the grid. In the absence of the project activity, the equivalent amount of power would have been supplied by the Indian grid, which is fed mainly by fossil fuel fired plants. Hence, the baseline for the project activity is the equivalent amount of power from the Indian Grid.

The combined margin ($EF_{grid,CM,y}$) is the result of a weighted average of two emission factor pertaining to the electricity system: the operating margin (OM) and build margin (BM), in accordance with the TOOL7: Tool to calculate the emission factor for an electricity system – Version 07 Calculations for this combined margin must be based on data from an official source (where available) and made publicly available. In India, Central Electricity Authority (CEA), Government of India provides this data, and accordingly the same has been used.

Calculations for this combined margin must be based on data from an official source (where available) and made publicly available. The CEA database version 19[23] is the latest available data & the same is considered for emission factor calculations.

The combined margin of the Indian grid used for the project activity is as follows:

| Parameter | Value | Nomenclature | Source |
|---|---|---|---|
| $EF_{grid,CM,y}$ | 0.9352 $tCO_2/MWh$ | Combined margin $CO_2$ emission factor for the | Calculated as the weighted average of the operating margin (0.75) & build margin (0.25) values, sourced from Baseline $CO_2$ |

[23] https://cea.nic.in/wp-content/uploads/baseline/2024/04/User_Guide__Version_19.0.pdf

| | | | |
|---|---|---|---|
| | | project electricity system in year y | Emission Database, Version 19.0, December-2023 published by Central Electricity Authority (CEA), Government of India. |
| $EF_{grid,OM,y}$ | 0.9580 $tCO_2/MWh$ | Operating margin $CO_2$ emission factor for the project electricity system in year y | Calculated as the last 3-year (2020-21, 2021-22, 2022-23) generation-weighted average, sourced from Baseline $CO_2$ Emission Database, Version 19.0, December-2023 published by Central Electricity Authority (CEA), Government of India. |
| $EF_{grid,BM,y}$ | 0.8670 $tCO_2/MWh$ | Build margin $CO_2$ emission factor for the project electricity system in year y | Baseline $CO_2$ Emission Database, Version 19.0, December-2023 published by Central Electricity Authority (CEA), Government of India. |

The default baseline for the grid connected solar power project has been defined appropriately in accordance with the applied methodology, ACM0002, version 22.0. The latest version (version 19) of the Central Electricity Authority (CEA) of India database has been used to determine the grid emission factor applicable for the project. The central electricity authority of India is the body established under section 3 of the Electricity (Supply) Act, 1948 of India. The functions and duties of CEA are delineated under Section 73 of the Electricity Act, 2003.[24]

## 3.5 Additionality

### 3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

[24] https://cea.nic.in/functions/?lang=en

☐ Yes ☒ No

### 3.5.2 Additionality Methods

The additionality of the Project shall be demonstrated by applying the following approach, consisting of two components:

(i) A Legal Requirement Test; and

(ii) An Additionality Test either based on a Positive List test or a projects-specific additionality test.

The project is not mandated/enforced by law and is entirely a voluntary activity. Since voluntary commitments/agreements within a sector or by an entity do not constitute the legal requirement.

| | |
|---|---|
| Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable). | This project applies an approved large-scale methodology ACM0002 "Consolidated baseline methodology for grid connected electricity generation from renewable sources", Version 22.0. Selected methodology has been applied together with the "tool to calculate the emission factor for an electricity system, version 7.0" and "tool for the demonstration and assessment of additionality, version 7.0.0". These are the latest version of the methodology and related additionality & emission factor calculation tool. |
| Describe how the proposed project meets the criteria for deemed additionality. | Project revenue is not financially attractive as discussed in investment analysis section below (benchmark and sensitivity analysis). Continuation of the current situation supply of equal amount of electricity by the newly built grid connected power plants. Continuation of the current situation is not considered as a realistic alternative due to increasing electricity demand therefore new power plants should be constructed which includes mainly thermal power plants. Implementation of the project is additional to the baseline scenario which is an alternative 2 above and therefore reduces the emissions.<br><br>The project activity comes under white category as per local regulation, thus there shall be no necessity of obtaining the Consent to Operate'' for White category of industries. Since project activity falls under white category and the non-polluting nature of |

| | |
|---|---|
| | project fulfils the compliance to the local laws and regulations.<br><br>The Project activity conforms to all the applicable laws and regulations in India:<br><br>1. Power generation using renewable energy is not a legal requirement or a mandatory option[25].<br>2. There are state and sectoral policies, framed primarily to encourage renewable power projects.<br>3. These policies have also been drafted realizing the extent of risks involved in the projects and to attract private investments.<br>4. The Indian Electricity Act, 2003 (May 2007 Amendment) does not influence the choice of fuel used for power generation.<br>5. There is no legal requirement on the choice of a particular technology for power generation.<br><br>The both alternatives are in compliance with laws and regulations required. There is no mandatory requirement to implement the project activity.<br><br>In accordance with common practice analysis there is no plants similar to the proposed project and built without carbon revenue, the proposed type of project should not be considered as a common practice. Hence, project is additional in this aspect. |

Additionality Assessment for large scale project activity -

The present project generates power using solar energy which is a renewable, zero emission source of energy. However, the project activity does not comply to the criteria of positive list as provided in CDM Tool 32: "Methodological Tool – Positive List of Technologies" and hence additionality of the project activity is demonstrated through a project specific additionality test.

Baseline considerations for the project are based on approved consolidated baseline methodology. ACM0002 (Version 22.0). The methodology requires the project investor to determine the additionality based on "Methodological Tool- Tool for the demonstration and

[25] Electricity Act. 2003 and its subsequent amendments

assessment of additionality", Version 7.0.0. The tool provides a step-wise approach to demonstrate and assess the additionality of a project (figure below). These steps are:

a) Step 0 Demonstration whether the proposed project activity is the first-of-its-kind;
b) Step 1 Identification of alternatives to the project activity;
c) Step 2 Investment analysis;
d) Step 3 Barriers analysis; and
e) Step 4 Common practice analysis

The step-wise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:

**Step 0: Demonstration whether the proposed project activity is the first-of-its-kind**

The project activity is a large-scale solar power project undertaken in Indian state of Madhya Pradesh. This is not the first such project to be installed in the country or in the state and therefore project activity does not meet this criterion.

**Step 1: Identification of alternatives to the project activity consistent with current laws and regulations**

**Sub-step 1a: Define alternatives to the project activity: Identify realistic and credible alternative(s) available to the project participants or similar project developers that provide outputs or services comparable with the proposed project activity.**

The purpose of the project activity is aimed to generate power from the solar power plant using renewable energy (solar) resources and feed the electricity generated to the grid. Hence, the following alternatives are considered:

**Alternative 1: The proposed project activity undertaken without being registered as a Proposed project activity.**

The option of proposed project activity being undertaken without being registered as a project activity is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, this is not a plausible option as deployment of the project activity faces significant financial barriers established in the subsequent section. The electricity generated from the project activity would have been fed to the India's National Grid, which has a number of power plants including existing and upcoming ones which would be used in the absence of project activity. In the absence of project activity, power from the same grid will continue to provide for the requirement of the country.

**Alternative 2: Continuation of the current situation (No proposed project activity and equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid).**

In such case the project owner would have continued without investment in project activity and continued with its usual business activities. In such case grid would continue with the fossil fuel-based power projects and this would result in GHG emissions. Hence, existing power plant and the new capacity add-on from a fossil fuel-based power plant is appropriate, realistic & credible baseline alternative for the project activity.

**Outcome of Sub-step 1a: Identified realistic and credible alternative scenario (s) to the project activity.**

Of the two alternatives outlined above, the first alternative is not possible as project activity is not viable without carbon credit benefits and second alternative is the baseline scenario for the project activity as per methodology. Therefore, continuation of the current situation is the likely alternative to the project activity.

The project being a green field project activity the baseline scenario in line with the methodology is “the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system Version 7.0”

**Sub-step 1b: Consistency with mandatory laws and regulations: The alternative(s) shall be in compliance with all applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g. to mitigate local air pollution.**

Installation of a large-scale solar power project including the one proposed as project activity is consistent with mandatory laws and regulations at national (India) and sub-national level (state of Madhya Pradesh). The environmental regulations, legislations and policy guidelines in respect to the project activity is governed by Ministry of Environment, Forest and Climate Change (MoEF&CC), Delhi supported by Central Pollution Control Board (CPCB). In accordance to Annexure-II MOEF&CC, OM on J-11013/41/2006-IA. II (I) dated on 07-july-2017 Solar Photovoltaic Power Projects are not covered under the ambit of EIA Notification, 2006 and its subsequent amendments and hence, it does not require preparation of EIA and pursuing Environmental Clearance from MoEF&CC.

Moreover, Solar Power Project is considered under “white category” for Consent to Establish/Operate as per regulation, and thus there is no necessity of obtaining the Consent to Operate’’. Since project activity falls under white category and the non-polluting in nature, the project fulfils the compliance to the local environmental laws and regulations.

Overall, the project activity is consistent with the following mandatory laws and regulations.

- Electricity Act, 2003 and its subsequent amendment as it does not influence the choice of fuel used for power generation nor has it mandated power generation using renewable energy only.
- Environmental (Protection) Act, 1986 and amendment(s)
- Environmental Impact Assessment (EIA) Notification, 2006 and amendment(s)
- The Air (Prevention and Control of Pollution) Act, 1981 including Rules 1982 and 1983 and amendment(s)
- The Water Prevention and Control of Pollution), Cess Act, 1977 including Rules 1978 and 1991 and amended in March, 2003
- Noise Pollution (Regulation and Control) Amendment Rules, 2017
- Hazardous and other Wastes (Management & Transboundary Movement) Rules, 2016 and amended in November, 2021
- E-waste (Management and Handling) Rules, 2020
- Bio-Medical Waste (Management and Handling) Rules 2016
- Plastics Waste Management Rules, 2016 amended in 2021
- Batteries (management and handling) rules 2019

**Outcome of Sub-step 1b**: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the enforcement in the region or country and EB decisions on national and/or sectoral policies and regulations.

Hence, both the alternatives enlisted above are found to comply with the mandatory laws and regulations taking into account the enforcement of the legislations in the region or country and EB decisions on national and/or sectoral policies and regulations. However, Alternative 2 "Continuation of the current situation" has been selected as the appropriate baseline alternative for this project activity in line with methodology.

**Step 2: Investment analysis**

Investment analysis is carried out to determine on whether the proposed project activity is economically or financially less attractive than at least one other alternative, identified in step 1, without the revenue from the sale of emission reductions credits. This is demonstrated in line with the Tools for sections as per "Investment Analysis" (Ver 14.0).

**Sub-step 2a: Determine appropriate analysis method**

There are three options for the determination of analysis method which are:

I. Simple Cost Analysis
II. Investment Comparison Analysis and
III. Benchmark Analysis

The Project activity envisages to export the power to Indian grid and the revenues from the sale of electricity would be generated in accordance with the terms and tariffs established in the Power Purchase Agreement (PPA). Thus, simple cost analysis (Option I) cannot be used as the analysis method as the sale of the units of generated electricity shall result in a revenue stream during the operations of the Project activity.

In the absence of the project activity grid electricity would have been the obvious choice for the Project which requires no investment. Hence investment comparison analysis (Option II) is also not appropriate for the project activity.

After eliminating Option, I and Option II, the use of Benchmark analysis (Option III) is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project activity for the investors, as well as provides a measure of the viability of the investment to generate revenues during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be employed for analysis of the said project.

**Sub-step 2b (Option III): Apply benchmark analysis**

The investment analysis using Benchmark analysis approach (Option III) has been chosen. Further, this method illustrates that the evaluation of the project by the PP was carried out before the decision to undertake the project was taken and management approval had been granted.

The chronology of key events associated with the projects which are important to define the date of investment date of the project are produced as below, the decision date of project implementation is 08-Decemebr-2021.

| S.I No. | Event | Date |
|---|---|---|

| | | |
|---|---|---|
| 1 | RFP (Request for Proposal)[26] for Development of Floating Solar Park at Omkareshwar Reservoir, Madhya Pradesh by RUMSL (Rewa Ultra Mega Solar Limited)[27] | 02-November-2021 |
| 2 | Investment Note formed by the board of M/s AMP Energy Green Seven Private Limited based on their market survey outcomes | 08-December-2021 |
| 3 | Bid submission date | 26-April-2022 |
| 4 | Reverse auction date | 06-May-2022 |
| 5 | Release of Letter of Award | 30-June-2022 |
| 6 | Power Purchase Agreement between M.P. Power Management Company Limited and RUMSL and AMP Energy Green Seven Private Limited | 04-August-2022 |
| 7 | First Work Order released for string combiner boxes | 03-November-2022 |
| 8 | Date of Commissioning | 06-June-2024 |

**Choice of Financial Indicator:**

According to the "Tool for demonstration and assessment of Additionality", Version 07.0.0, the financial indicator can be based either on (1) project IRR or (2) equity IRR. There is no general preference between the approaches (1) or (2). The benchmark chosen for analysis shall be fully consistent with the choice of approach. Therefore, in accordance with the guidance, the relevant financial indicator for project activity has been chosen as post tax equity IRR.

**Default Value Benchmark:**

The cost of equity is determined by selecting the values provided in the Appendix, i.e., Default values for cost of equity (expected return on equity) is presented below:

Appendix of the CDM TOOL 27: Investment Analysis Version 11.0 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in India = 10.55%

The Required return on equity (benchmark) was computed in the following manner:

**Nominal Benchmark = {(1 + Real Benchmark) * (1 + Inflation rate)} – 1**

Where:

Default value for Real Benchmark = 10.55% (as per Appendix of CDM TOOL 27: Investment Analysis, Ver 11.0).

Inflation Rate forecast for by Reserve Bank of India (RBI) (i.e. Central Bank of India) for India.

**Benchmark estimation:**

Benchmark for Floating Solar project of amp Energy

[26] https://jmkresearch.com/wp-content/uploads/2021/11/RUMSL-600-MW-Floating-Solar-Madhya-Pradesh-Nov-2021.pdf

[27] http://rumsl.mp.gov.in/

| Description | Value | Source Link |
|---|---|---|
| Default Value for India as per UNFCCC guidelines | 10.55% | https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v11.0.pdf<br>Dated -29/October/2021 |
| **Inflation forecast as per RBI for 10 yr** | 4.00% | https://www.rbi.org.in/Scripts/BS_PressReleaseDisplay.aspx?prid=52450<br>Dated -22/October/2021 |
| Benchmark | 14.97% | Calculated |
| Decision date | 08-December-2021 | |

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III): The Post tax Equity IRR is evaluated for the entire lifetime of the project activity, i.e. 25 years. It is calculated based on the cash outflows from and cash inflows into the project activity. substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark). Thus, it can be easily concluded that project activity is additional & is not business as usual scenario.

The input assumption referred below:

| Details of the project | | Source | Link |
|---|---|---|---|
| Project Location | Madhya Pradesh | Investment Note formed by the board of M/s AMP Energy Green Seven Private Limited | Page no 01 |
| Total Capacity (MWac) | 100.00 | Investment Note | - |
| Expected Date of Commissioning | 31-October-2023 | PPA [Section 4.1(b) Page-20] | - |
| Life of the plant (Yrs.) | 25 | Investment Note | - |
| **Generation of electricity** | | | |
| PLF (%) | 24.40% | Investment Note | - |
| Annual generation (kWh) | 213,744,000 | Calculated Value | - |
| Annual Degradation per year from 2nd Yr onwards | 0.30% | As per technical specification report | - |
| Tariff rate at the decision making (INR/kWh) | 3.21 | Investment Note | - |
| **Operation and maintenance cost and Insurance** | | | |
| O & M Expenses (INR Mn.) | 25.96 | Investment Note | - |
| O & M free for (Yr.) | - | | - |

| | | | |
|---|---|---|---|
| Escalation in the operational expenses (%) | 2.00% | Investment Note | - |
| Admin Fee (₹ mn) | 0.70 | Investment Note | - |
| Escalation (%) | 4.00% | Investment Note | - |
| Salary (₹ mn) | 0.50 | Investment Note | - |
| Escalation (%) | 4.00% | Investment Note | - |
| Lender maintenance (₹ mn) | 0.45 | Investment Note | - |
| Escalation (%) | 4.00% | Investment Note | - |
| Insurance (₹ mn) | 8.85 | Investment Note | - |
| Escalation (%) | 0.00% | | - |
| Any other cost | 0.00 | Investment Note | - |
| **Financial parameters** | | | |
| TOTAL COST (INR Mn.) | 6,000.00 | Investment Note | |
| Debt Ratio (%) | 75% | Investment Note | |
| Equity Ratio (%) | 25% | Investment Note | |
| Loan Amount (INR Mn.) | 4,500.00 | Calculated Amount | |
| Equity Investment (INR Mn.) | 1,500.00 | | |
| **Term loan** | | | |
| Loan Amount (INR Mn.) | 4,500.00 | 75:25 ratio - Investment Note | - |
| Interest rate (%) | 9.15% | Investment Note | - |
| Loan Tenure (Qtr.) | 80.00 | Investment Note | - |
| Moratorium Period (Qtr.) | 4.00 | Investment Note | - |
| Repayment Period (Qtr.) | 76.00 | Calculated Value | - |
| Repayment instalments value (INR Mn.) | 59.21 | Calculated Value | - |
| 1st instalment from (Qtr. end) | 31-Dec-24 | Considered from the next Quarter End | - |
| **Book Depreciation (SLM Method)** | | | |
| Gross Depreciable Value (INR Mn.) | 6,000.00 | Calculated Value | |
| Book depreciation % | 4% | Calculated value | |
| Salvage Value (%) | 10.00% | DPR | - |
| Salvage value (INR Mn.) | 600.00 | Calculated Value | |
| Net Depreciable Value (INR Mn.) | 5,400.00 | Calculated Value | |
| **IT Depreciation** | | | |
| IT Depreciation (%) | 40.00% | As Per Depreciation Rates for Power Generating Units | https://incometaxindia.gov.in/charts%20%20tables/depreciation%20rates.htm |

| **Income Tax** | | | |
|---|---|---|---|
| Income tax rate (%) | 30.00% | Tax rated applicable @ FY 2020-21 for a domestic company | https://incometaxindia.gov.in/Booklets%20%20Pamphlets/e-Pdf-INCOME-TAX-RATES-english-2022.pdf |
| MAT (%) | 15.00% | | |
| Surcharge (%) | 7.00% | | |
| Health & Education cess (%) | 4.00% | | |
| Service tax (%) | 18.00% | As Per Income Tax Rule | https://old.cbic.gov.in/resources//htdocs-cbec/gst/Work%20Contracts%20in%20GST.pdf |
| **Final Tax rates** | | | |
| Income tax rate (%) | 33.38% | Calculated Value | |
| MAT (%) | 16.69% | Calculated Value | |
| Service tax (%) | 18.72% | Calculated Value | |

**Outcome of Step 2:**

Calculation of equity IRR is submitted in the form of excel sheet.

| Equity IRR | Benchmark (Equity IRR) |
|---|---|
| 7.27 % | 14.97% |

This substantiates that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark Equity IRR). Thus, it can be easily concluded that project activity is additional & is not business as usual scenario.

**Sub-step 2d: Sensitivity Analysis**

Addressing Guidance under Sensitivity Analysis Version-14.0, following factors has been subjected to sensitivity analysis:

1. PLF
2. O&M Cost
3. Project Cost
4. Tariff

The rationale of sensitivity is, "The ultimate objective of the sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the scenario presented, in order to provide a cross-check on the suitability of the assumptions used in the development of the investment analysis."

**Sensitivity Analysis**

The results of sensitivity analysis show that the even with a variation of +10% & -10% in the project cost, O&M cost, PLF and Tariff Rate Equity IRR is significantly lower than the benchmark. And it is evident from the results given above; the project remains additional even under the most favourable conditions.

| Probability to breach the benchmark: |
|---|

| **Sensitivity Parameter 1: PLF or generation** |
|---|
| The PLF or generation is considered at the 90% probability level for entire lifetime of the project activity. Hence, there is no probability to get variation in the same. However, Sensitivity is carried out for +/-10% even then the benchmark is not breached.<br>Actual PLF generation achieved by the solar plant since commissioning and found that the average PLF achieved is approximately 18.48%, which is less than the estimated PLF 24.40%. Sensitivity analysis with a 10% variation with PLF will not breach the benchmark. The breaching value for the PLF will be +20.08% (PLF=29.30%) which is unlikely scenario. The result of the sensitivity analysis is mentioned in the below table. |
| **Sensitivity Parameter 2: O&M** |
| The sensitivity analysis reveals that O&M will breach the benchmark at negative values (-347%) and is hypothetical case. Since the O&M cost is generally subject to escalation and also subject to inflationary pressure, any reduction in the O&M costs is highly unlikely. Hence, the reduction in the O&M cost is highly unlikely. The actual investment of INR 6,145.30 million INR as per the CA certificate is 2% higher than the estimated cost (6,000.0 million INR). The estimated cost is considered as conservative. |
| **Sensitivity Parameter 3: Project Cost** |
| Project Cost for financial analysis is considered based on the information of DPR, being available at the time of investment decision making to go ahead with the project activity. However, the Sensitivity is carried out for 10% variation and for threshold level below which benchmark is not breached. Thus, it is unlikely that project cost will change beyond sensitivity range.<br>The actual investment of INR 6,145.30 million INR as per the CA certificate is 2% higher than the estimated cost (6,000.0 million INR). The estimated cost is considered as conservative. |
| **Sensitivity Parameter 4: Tariff Rate** |
| The tariff is fixed for entire lifetime of the project activity. Hence, there is no probability to get variation for the same. However, Sensitivity is carried out for +/-10% even then the benchmark is not breached.<br>The tariff base rate is based on Investment note, which was available at the time of investment decision. This is also crosschecked through the actual invoices raised to M.P. POWER MANAGEMENT COMPANY LIMITED (MPPMCL) and found to be similar as 3.21 Rs/kWh. The PPA is fixed without any escalation for 25 years. Hence, the tariff considered in the investment analysis is appropriate and will not breach the benchmark in future. |

**Result of Sensitivity Analysis Table:**

| Parameters | -10% | IRR (Base Case) | +10% | IRR Breaching Value |
|---|---|---|---|---|
| PLF or Generation | 3.79% | 7.27% | 10.91% | 20.08% |
| O&M Cost | 7.47% | 7.27% | 7.02% | -347.00% |
| Project Cost | 10.86% | 7.27% | 4.41% | -18.50% |
| Tariff Rate | 3.79% | 7.27% | 10.91% | 20.08% |

**Step 3: Barrier Analysis**

As per Tool for demonstration and assessment of additionality" (Version 07.0.0), Step 2 or Step 3 or both can be used to demonstrate additionality of the project activity. In this case, Step 3 is not being used for the purpose.

**Step 4: Common Practice Analysis**

As per para 57 of "Tool for demonstration and assessment of additionality" (Version 07.0.0), Step 2 analysis shall be complemented with an analysis of extent to which the proposed project type (e.g., technology or practice) has already diffused in the relevant sector and region. This test is a credibility check to complement the investment analysis.

Common Practice Analysis – 100 MW Floating Solar Power Project by AMP Energy In Madhya Pradesh

**Step (1):** Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

| Range | Capacity | Unit |
|---|---|---|
| +50% | 150 | MW |
| Capacity of the proposed project activity | 100 | MW |
| -50% | 50 | MW |

**Step (2):** Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

- The projects are located in the applicable geographical area;
- The projects apply the same measure as the proposed project activity;
- The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;
- The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g., clinker) as the proposed project plant;
- The capacity or output of the projects is within the applicable capacity range for the chosen projects.
- The projects started commercial operation before the Project design document is published for public comment period or before the start date of proposed project activity, whichever is earlier for the proposed project activity.

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step

(2) as follows:

- As the project is in Madhya Pradesh state of India, therefore, projects in the geographical area of Madhya Pradesh have been chosen for analysis. The project activity involves generation of electricity from solar energy. The project activity is in the states of Madhya Pradesh in India and the policy applicable for the solar projects is regulated by respective state policy. The policies/tariff for each state is regulated by State Electricity Regulatory Commissions of respective states and they differ for respective states. The project is Floating PV solar and a 25% equity-based project and generated power will be exported to grid. Hence, all state electricity regulations are applicable for this project. So, solar projects, which are exporting electricity to grid are followed different state electricity regulatory policies.

- The project activity is a green-field solar power project and uses measure (b) “Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies”. Therefore, projects applying same measure (b) are candidates for similar projects.
- The energy source used by the project activity is solar. Hence, only solar energy projects have been considered for analysis.
- The project activity produces electricity; therefore, all power plants that produce electricity are candidates for similar projects.
- The capacity range of the projects is within the applicable capacity range from 50 MW to 150 MW.
- First Purchase Order date is 03-November-2022 (PO for the purchase of string combiner boxes) for this project activity. Therefore, projects which have started commercial operation before 03-November-2022 have been considered for analysis.

PP has considered the MP state of India for CPA and checked all the floating solar projects from CEA database (https://cea.nic.in/old/reports/others/planning/rpm/Plant-wise%20details%20of%20RE%20Installed%20Capacity-merged.pdf) and SECI data (https://www.seci.co.in/ists/solar) that falls within the specified range.

Numbers of similar projects identified, which fulfil above-mentioned condition are considered from the Central Electricity Authority (Plant wise details of All India Renewable Energy Projects). The number of projects following all the conditions met above, as identified above are $N_{solar}$ = 0[28].

**Step (3):** Within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number, $N_{all}$.

Project activities, which have got registered or are under validation with CDM/VCS/GS/GCC have been excluded in this step. The list of the power plants identified is provided to the Verifier. After excluding the registered and under validation projects the total number of projects.

$N_{all}$ = 0

**Step (4):** Within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number $N_{diff}$.

$N_{diff}$ = 0

**Step (5):** calculate factor $F=1-N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology

28 https://cea.nic.in/wp-content/uploads/rpm_division/2024/02/Report_On_Under_Construction_Renewable_Energy_Projects_for__the_Month_December_2023.pdf

used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

The factor for the project activity is calculated as follows:

In terms of the formula: $F=1-(N_{diff}/N_{all})$, the result is mathematically meaningless.

$N_{all} - N_{diff} = 0$.

i.e. $N_{all} = N_{diff}$

So, we can write, $F=1-(N_{diff}/N_{all}) = 1-(N_{all}/N_{all}) = 1-1 = 0$.

As per methodological tool "Common practice" version 03.1, the proposed project activity is a "common practice" within a sector in the applicable geographical area if the factor F is greater than 0.2 and $N_{all} - N_{diff}$ is greater than 3. Thus, if both conditions are fulfilled, then project activity will be a common practice. Otherwise, the project activity is treated as not a common practice.

Outcome of Step 5:

Thus, the project does not fulfil the requirement that F is greater than 0.2 and $N_{all}-N_{diff}$ is greater than 3, and the project is not a common practice.

## 3.6 Methodology Deviations

There is no deviation in methodology applied by the project activity hence not applicable.

# 4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

## 4.1 Baseline Emissions

The baseline emissions are the product of generated power from the solar power plant baseline $EG_{PJ,Y}$ expressed in MWh of electricity produced by the renewable generating unit multiplied by an emission factor. As per para 57 of Methodology ACM0002 Ver-22.0 the baseline emission equations are used.

$BE_y = EG_{PJ,Y} * EF_{Grid,CM,y}$ Equation (11)

Where:
$BEy$ = Baseline emissions in year y ($tCO_2e/yr$)
$EG_{PJ,Y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{Grid,CM,y}$ = Combined margin $CO_2$ emission factor for grid connected power generation in year y calculated using TOOL07 ($tCO_2e/MWh$)

As per methodology, combined grid emission factor as per the “Tool to calculate the emission factor for an electricity system” version 07 is calculated as below.

$CO_2$ Baseline Database for the Indian Power Sector, Version 19.0, December - 2023 published by Central Electricity Authority (CEA), Government of India has been used for the calculation of emission reduction.
As per Methodological tool: Tool to calculate the emission factor for an electricity system (Version 7.0, EB 100, Annex 4), following six steps have been followed:

a) **Step 1:** Identify the relevant electricity systems;
b) **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional);
c) **Step 3:** Select a method to determine the operating margin (OM);
d) **Step 4:** Calculate the operating margin emission factor according to the selected method;
e) **Step 5:** Calculate the build margin (BM) emission factor;
f) **Step 6:** Calculate the combined margin (CM) emission factor.

**Step 1: Identify the relevant electricity systems**

As described in tool “For determining the electricity emission factors, identify the relevant project electricity system. Similarly, identify any connected electricity systems”. It also states

That “If the DNA of the host country has published a delineation of the project electricity system and connected electricity systems, these delineations should be used”. Keeping this into consideration, the Central Electricity Authority (CEA), Government of India have Indian grid.

However, since August 2006, however, all regional grids except the Southern Grid had been integrated and were operating in synchronous mode, i.e. at same frequency. Consequently, the Northern, Eastern, Western and North-Eastern grids were treated as a single grid named as Unified Indian Grid from FY 2007-08 onwards for the purpose of this $CO_2$ Baseline Database. As of 31 December 2013, the Southern grid has also been synchronized with the Unified Indian Grid, hence forming one unified Indian Grid. Since the project supplies electricity to the Indian grid, emissions generated due to the electricity generated by the Indian grid as per CM calculations will serve as the baseline for this project.

**Table: Geographical Scope of Indian Electricity Grid**

| Northern | Eastern | Western | North-Eastern | Southern |
|---|---|---|---|---|
| Chandigarh | Bihar | Chhattisgarh | Arunachal Pradesh | Andhra Pradesh |

| | | | | |
|---|---|---|---|---|
| Delhi | Jharkhand | Gujarat | Assam | Karnataka |
| Haryana | Orissa | Daman & Diu | Manipur | Kerala |
| Himachal Pradesh | West Bengal | Dadar & Nagar Haveli | Meghalaya | Tamil Nadu |
| Jammu & Kashmir | Sikkim | Madhya Pradesh | Mizoram | Telangana |
| Punjab | Andaman & Nicobar | Maharashtra | Nagaland | Puducherry |
| Rajasthan | | Goa | Tripura | Lakshadweep |
| Uttar Pradesh | | | | |
| Uttarakhand | | | | |

**Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)**

The project owner may choose between the following two options to calculate the operating margin and build margin emission factor:

**Option I:** Only grid power plants are included in the calculation.

**Option II:** Both grid power plants and off-grid power plants are included in the calculation.

The project owner has chosen only grid power plants in the calculation.

**Step 3: Select a method to determine the operating margin (OM)**

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods, which are described under Step 4:

(a) Simple OM; or

(b) Simple adjusted OM; or

(c) Dispatch data analysis OM; or

(d) Average OM.

The data required to calculate Simple adjusted OM and Dispatch data analysis OM is not possible due to lack of availability of data to project developers. The choice of other two options for calculating operating margin emission factor depends on generation of electricity from low-cost/ must-run sources. In the context of the methodology low cost/must run resources typically include hydro, geothermal, solar, low-cost biomass, nuclear and solar generation.

**Share of Must-Run (Hydro/Nuclear) (% of Net Generation)**

| | 2017-18 | 2018-19 | 2019-20 | 2020-21 | 2021-22 | 2022-23 |
|---|---|---|---|---|---|---|
| India | 17.0% | 16.5% | 17.0% | 16.5% | 15.8% | 15.3% |

Data Source: Central Electricity Authority (CEA) database Version 19.0, December 2023[29]

The above data clearly shows that the percentage of total grid generation by low-cost/ must-run plants (on the basis of average of five most recent years) for the Indian grid is less than 50 % of the total generation. Thus, the Average OM method cannot be applied, as low cost/must run resources constitute less than 50% of total grid generation.

The simple OM emission factor is calculated as the generation-weighted average $CO_2$ emissions per unit net electricity generation ($tCO_2e/MWh$) of all generating power plants serving the system, not including low-cost/must-run power plants/units.

For the simple OM, the simple adjusted OM and the average OM, the emissions factor can be calculated using either of the two following data vintages:

(a) **Ex-ante option:** if the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

**OR**

(b) **Ex-post option:** if the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring.

Project Owner has chosen ex-ante option for calculation of Simple OM emission factor using a 3-year generation-weighted average, based on the most recent data available at the time of submission of the PD to the VVB for validation.

OM determined at validation stage will be the same throughout the crediting period. There will be no requirement to monitor & recalculate the emission factor during the crediting period.

**Step 4: Calculate the operating margin emission factor ($EF_{grid,OMSimple,y}$) according to the selected method**

The operating margin emission factor has been calculated using a 3-year data vintage

| Net Generation in Operating Margin (GWH) (incl. Imports) | **2020-21** | **2021-22** | **2022-23** |
|---|---|---|---|
| INDIAN Grid | 9,58,218 | 10,35,672 | 11,17,846 |

| Simple Operating Margin ($tCO_2e/MWh$) (incl. Imports) | **2020-21** | **2021-22** | **2022-23** |
|---|---|---|---|
| INDIAN Grid | 0.9402 | 0.9605 | 0.9710 |

| **Weighted Generation Operating Margin** | |
|---|---|
| INDIAN Grid | 0.9580 |

[29] https://cea.nic.in/wp-content/uploads/baseline/2024/01/User_Guide__Version_19.0.pdf

**Step 5: Calculate the build margin (BM) emission factor ($EF_{grid,BM,y}$)**

As per Methodological tool: "Tool to calculate the emission factor for an electricity system" (Version 7.0, EB 100, Annex 4) para 72:

In terms of vintage of data, the project owner can choose between one of the following two options:

(a) **Option 1** - for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of PD submission to the VVB for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the VVB. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

(b) **Option 2** - For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

Option 1 as described above is chosen by Project Owner to calculate the build margin emission factor for the project activity. BM is calculated ex-ante based on the most recent information available at the time of submission of PD and is fixed for the entire crediting period.

| **Build Margin ($tCO_2e/MWh$) (not adjusted for imports)** | |
|---|---|
| | 2022-23 |
| INDIAN Grid | 0.8670 |

**Step 6: Calculate the combined margin (CM) emission factor ($EF_{grid,CM,y}$)**

As per Methodological tool: "Tool to calculate the emission factor for an electricity system" (Version 7.0, EB 100, Annex 4) para 81:

The calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

(a) Weighted average CM; or
(b) Simplified CM.

Project Owner has chosen option (a) i.e weighted average CM to calculate the combined margin emission factor for the project activity.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} * W_{OM} + EF_{grid,BM,y} * W_{BM}$$

Where:

| | |
|---|---|
| $EF_{grid,BM,y}$ | = Build margin $CO_2$ emission factor in year y ($tCO_2e/MWh$) |
| $EF_{grid,OM,y}$ | = Operating margin $CO_2$ emission factor in year y ($tCO_2e/MWh$) |
| $W_{OM}$ | = Weighting of operating margin emissions factor (per cent) |
| $W_{BM}$ | = Weighting of build margin emissions factor (per cent) |

The following default values should be used for $W_{OM}$ and $W_{BM}$:

For solar project activities: $W_{OM}$ = 0.75 and $W_{BM}$ = 0.25 (owing to their intermittent and non-dispatchable nature) for the first crediting period and for subsequent crediting periods. Since project activity is of power generation by using solar, the above weightage has been considered for OM and BM.

Therefore, $EF_{grid,CM,y}$ = 0.9580 * 0.75 + 0.8670 * 0.25
= 0.9352 $tCO_2e/MWh$

For Green field project

$$EG_{PJ,y} = EG_{Facility,y}$$

Where;

| | |
|---|---|
| $EG_{PJ,y}$ | = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr) |
| $EG_{Facility,y}$ | = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr) |

$BE_y$ = 211,556 * 0.9352

**$BE_y$ = 197,846 $tCO_2e$**

## 4.2 Project Emissions

As per para 40 of the applied methodology ACM0002 (version 22.0), for most renewable energy power generation project activities, $PE_y$ = 0
Since, this is a renewable energy power project, the project emissions are considered as zero. Hence, project emissions $PE_y$ = 0 $tCO_2e$.

## 4.3 Leakage Emissions

As per Para 71 of ACM0002 Version 22.0 The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc. are neglected. No other leakage emissions are considered.

## 4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

As per the para 72 of ACM0002 version 22.0, the formula to calculate the emission reductions is

$ER_y = BE_y - PE_y$ **Equation (17)**

Where:

$ER_y$ = Emission reductions in year y ($tCO_2e/yr$)

$BE_y$ = Baseline emissions in year y ($t\ CO_2e/yr$)

$BE_y$ = Project emissions in year y ($t\ CO_2e/yr$)

Therefore, Net GHG Emission Reductions and Removals are calculated as follows:

| Vintage period | Estimated baseline emissions ($tCO_2e$) | Estimated project emissions ($tCO_2e$) | Estimated leakage emissions ($tCO_2e$) | Estimated reduction VCUs ($tCO_2e$) | Estimated removal VCUs ($tCO_2e$) | Estimated total VCUs ($tCO_2e$) |
|---|---|---|---|---|---|---|
| 06-June-2024 to 31-December-2024 | 114,459 | 0 | 0 | 114,459 | 0 | 114,459 |
| 01-January-2025 to 31-Decemebr-2025 | 199,292 | 0 | 0 | 199,292 | 0 | 199,292 |
| 01-January-2026 to 31-December -2026 | 198,694 | 0 | 0 | 198,694 | 0 | 198,694 |
| 01-January-2027 to 31-Decemebr-2027 | 198,097 | 0 | 0 | 198,097 | 0 | 198,097 |
| 01-Jan-2028 to 31-Dec-2028 | 197,503 | 0 | 0 | 197,503 | 0 | 197,503 |
| 01-January-2029 to 31-Decemebr-2029 | 196,910 | 0 | 0 | 196,910 | 0 | 196,910 |
| 01-January-2030 to 31-December-2030 | 196,319 | 0 | 0 | 196,319 | 0 | 196,319 |
| 01-January-2031 to 05-June-2031 | 83,653 | 0 | 0 | 83,653 | 0 | 83,653 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Total | 1,384,927 | 0 | 0 | 1,384,927 | 0 | 1,384,927 |

This project activity doesn't require the assessment of permanence risk along with the proportional split of buffer pool allocation between the estimated reductions & removals.

Hence, following the PD filling guidelines of Version 4.3, the table for assessing the permanence risk & the table for buffer pool allocation are not included in this section.

# 5 MONITORING

## 5.1 Data and Parameters Available at Validation

| Data / Parameter | $EF_{grid,OM,y}$ |
|---|---|
| Data unit | $tCO_2e/MWh$ |
| Description | Operating Margin $CO_2$ emission factor in year y |
| Source of data | $CO_2$ Emission Database, Version 19.0, December - 2023 published by Central Electricity Authority (CEA), Government of India. |
| Value applied | 0.9580 |
| Justification of choice of data or description of measurement methods and procedures applied | Operating margin $CO_2$ emission factor for the project electricity system in year y Calculated as the last 3-year (2020-21, 2021-2022 & 2022-23) generation-weighted average, sourced from Baseline $CO_2$ Emission Database, Version 19.0, December - 2023 published by Central Electricity Authority (CEA), Government of India |
| Purpose of data | For the calculation of the Baseline Emission |
| Comments | - |

| Data / Parameter | $EF_{grid,BM,y}$ |
|---|---|
| Data unit | $tCO_2e/MWh$ |
| Description | Build Margin $CO_2$ emission factor in year y |

| | |
|---|---|
| Source of data | $CO_2$ Emission Database, Version 19.0, December - 2023 published by Central Electricity Authority (CEA), Government of India. |
| Value applied | 0.8670 |
| Justification of choice of data or description of measurement methods and procedures applied | Build margin $CO_2$ emission factor for the project electricity system in year y Baseline $CO_2$ Emission Database, Version 19.0. December - 2023 published by Central Electricity Authority (CEA), Government of India |
| Purpose of Data | For the calculation of the Baseline Emission |
| Comments | - |

| | |
|---|---|
| Data / Parameter | $EF_{grid,CM,y}$ |
| Data unit | $tCO_2e/MWh$ |
| Description | Combined Margin $CO_2$ emission factor in year y |
| Source of data | Combined margin $CO_2$ emission factor for the project electricity system in year y Calculated as the weighted average of the operating margin (0.75) & build margin (0.25) values, sourced from Baseline $CO_2$ Emission Database, Version 19.0, December - 2023 published by Central Electricity Authority (CEA), Government of India |
| Value applied | 0.9352 |
| Justification of choice of data or description of measurement methods and procedures applied | The combined margin emissions factor is calculated as follows:<br>$EF_{grid,CM,y} = EF_{grid,OM,y} * W_{OM} + EF_{grid,BM,y} * W_{BM}$<br>Where:<br>$EF_{grid,BM,y}$= Build margin $CO_2$ emission factor in year *y* ($tCO_2/MWh$)<br>$EF_{grid,OM,y}$= Operating margin $CO_2$ emission factor in year *y* ($tCO_2/MWh$)<br>$W_{OM}$ = Weighting of operating margin emissions factor (%) = 75%<br>$W_{BM}$= Weighting of build margin emissions factor (%) = 25% |
| Purpose of Data | For the calculation of the Baseline Emission |

| | |
|---|---|
| Comments | - |

## 5.2 Data and Parameters Monitored

<table>
<tr><td>Data / Parameter</td><td>$EG_{Facility,y}$</td></tr>
<tr><td>Data unit</td><td>MWh/year</td></tr>
<tr><td>Description</td><td>Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)</td></tr>
<tr><td>Source of data</td><td>Monthly Joint Meter Readings</td></tr>
<tr><td>Description of measurement methods and procedures to be applied</td><td>Once in a month, the meter readings measurement at metering point is carried out by state electricity board officials in the presence of O&M officials/PP representatives. The calculations/ measurement of net electricity supplied to grid is under purview of state electricity board and the PP has no control on the same.</td></tr>
<tr><td>Frequency of monitoring/recording</td><td>Continuous monitoring & monthly recording</td></tr>
<tr><td>Value applied</td><td>211,556 (Estimated)</td></tr>
<tr><td>Monitoring equipment</td><td>
<table>
<tr><td>Type of meter</td><td>ABT bidirectional tri-vector compliant energy meters</td></tr>
<tr><td>Location of meter</td><td>All meters are located at the Interconnection point/Substation of the solar power plant. The joint meter readings are taken every month from both these meters which form the basis to raise monthly invoice.</td></tr>
<tr><td>Accuracy of meter</td><td>0.2S</td></tr>
<tr><td>Serial number of meters</td><td>Main Meter:<br>(1) Q0944018 (2) Q0944019<br>(3) Q0944020 (4) Q0944021<br>Check Meters:<br>(1) Q0944022 (2) Q0944023<br>(3) Q0944024 (4) Q0944025,</td></tr>
<tr><td>Calibration frequency</td><td>As per MADHYA PRADESH ELECTRICITY DISTRIBUTION CODE (MPERC) (Installation and Operation of Meters) (Amendment) Regulations, 2019[30], all Interface Meters shall be tested on-site</td></tr>
</table>
</td></tr>
</table>

[30] https://mperc.in/uploads/regulation_document/bf79dc4ee5f3f9b84255278af85ba3d3.11

| | |
|---|---|
| | <table><tr><td></td><td>using accredited test laboratory for routine accuracy testing at least once in a year and recalibrated if required.</td></tr><tr><td>Date of Calibration/ validity</td><td>Since the newly installed energy meters are pre-calibrated by default, the test was also conducted to verify their accuracy for the new installed meters on 31-january-2024 by the OEM (Validity is up to 1 years) for all the 8 meters. Hence, the date of next calibration of the meters to be conducted before 30-January-2025.</td></tr><tr><td>Calibration Status</td><td>The energy meters were tested during the commissioning and considering the calibration dates mentioned above, the calibration of the meters is still valid and calibration will be done during verification of this project activity.</td></tr></table> |
| QA/QC procedures to be applied | According to Central Electricity Authority (Installation and Operation of meters) Regulations 2019[31], all the electricity meters (interface & consumer meters) shall be tested at least once in five years and recalibrated, if required.<br><br>Hence, all the meters that are part of this project activity will be tested at least once in 5 years and recalibrated, if required. |
| Purpose of data | Calculation of baseline emissions |
| Calculation method | Electricity exported/imported to the grid is in kWh. However, for the calculation purpose electricity exported is converted in MWh.<br><br>The Bi directional energy meter measures both export and import of project activity. The Net electricity supplied to the grid by the project activity will be calculated as a difference of electricity exported to the grid, electricity imported from the grid obtained from joint meter reading certificates/credit notes issued by state electricity board as per below equation:<br><br>$EG_{facility,y} = EG_{Export} - EG_{Import}$<br><br>The joint reading at metering point is carried out once in a month in presence of O&M officials and state electricity board personnel. The calculations/measurement of net electricity supplied to grid is under purview of state electricity board and the project owner |

[31] https://cea.nic.in/wp-content/uploads/2020/02/CEA_metering_regulation_amendment_2019.pdf

| | |
|---|---|
| | has no role on it. Project owner will get value of net electricity supplied to grid and hence this parameter is mentioned as a part of monitoring plan.<br><br>Cross Checking: Quantity of net electricity supplied to the grid will be cross checked from the invoices raised to the State Electricity Board. |
| Comments | Data will be archived electronically for a period of 2 years beyond the end of crediting period. |

## 5.3 Monitoring Plan

The monitoring plan is developed in accordance with the modalities and procedures for the project activities and is proposed for grid-connected solar power project being implemented. The monitoring plan, which will be implemented by the project proponent describes about the monitoring organization, parameters to be monitored, monitoring practices, quality assurance, quality control procedures, data storage and archiving. The authority and responsibility for registration, monitoring, measurement, reporting and reviewing of the data rests with the project owner. The project owner proposed the following structure for data monitoring, collection, data archiving and calibration of equipment's for this project activity.

**Module Cleaning**

Cleaning of modules is very important for achieving desired yield and performance ratio in solar power project. It is observed that many times, the system performance is inferior just because the PV modules have not been cleaned properly and if they are clean, Dirt and bird litter excreta are the two major factors which affects the module output.

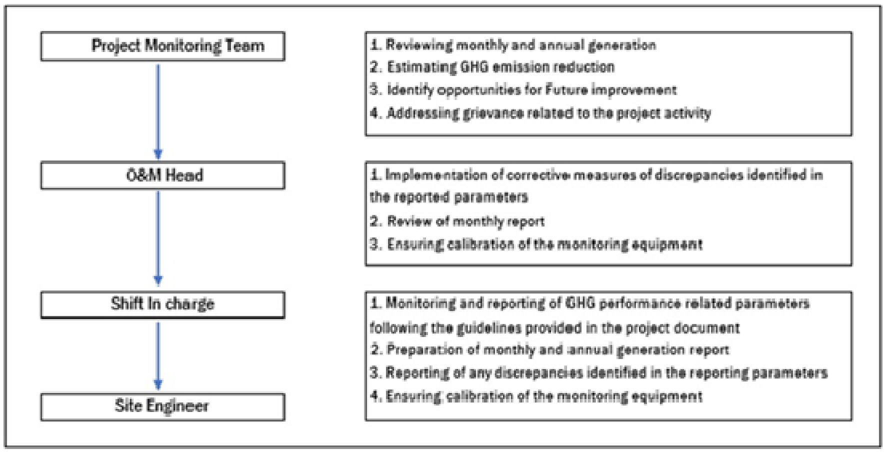
Fresh water (TDS < 1500 mg/L) will be used to clean the modules. If needed, a mild, nonabrasive, non- caustic detergent with a final fresh water and detergent solution mix between 6.5 < pH < 8.5 at 25°C will be used.

The water used will not be too cold as well as not too hot. Its temperature should be Luke warm water. The water used will be free from dirt and mud. Typically, the water used will be de-mineralized water so that, salt preposition of module surface will not take place. Salty water may also lead to corrosion on module frame as well as the mounting structures. The cleaning wiper used will be non-adhesive and the water should be removed from top to bottom approach.

**Data Measurement**

The export and import energy will be measured continuously using above mentioned Main and Check meters located at the substations. Readings of meters shall be taken on monthly basis by authorized officers of DISCOM in the presence of PP or representative of PP. Based on the Meter Reading Statement to PP, invoices will be raised. These invoices can be used for cross checking the meter readings taken for the respective project activity.

**Organization structure with role and responsibility**




<u>Data collection and archiving</u>

Readings from meters will be collected in the presence of the plant in-charge. Export and Import data would be recorded and stored in logs as well as in electronic form on a daily basis. The records are checked periodically by the Plant Manager and discussed thoroughly with the plant supervisor. The period of storage of the monitored data will be 2 years after the end of crediting period or till the last issuance of VCUs for the project activity whichever occurs later.

<u>Emergency preparedness</u>

The project activity will not result in any unidentified activity that can result in substantial emissions from the project activity. No need for emergency preparedness in data monitoring is visualized. In the event that the main meter, which is used to record the net electricity exported by the project, is found to be faulty it will be repaired or replaced and the data from the check meter will be used in its place. In the unlikely event that the check meter fails it will also be repaired or replaced.

<u>Personnel training</u>

In order to ensure a proper functioning of the project activity and a properly monitoring of emission reductions, the staff are made trained. The plant helpers are trained in equipment operation, data recording, reports writing, operation and maintenance and emergency procedures in compliance with the monitoring plan.

<u>QA/QC procedures</u>

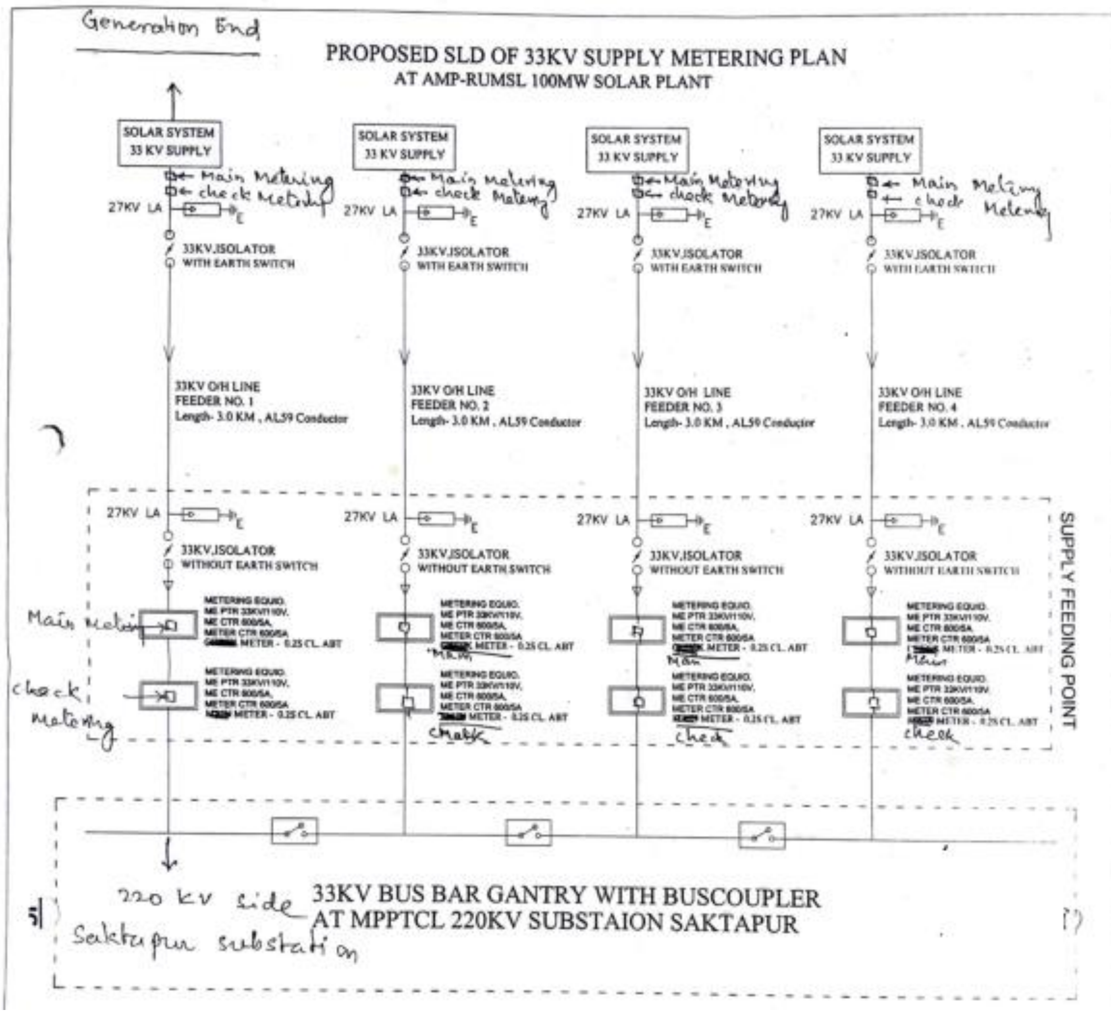
The energy meters at the feeders are maintained and owned by state electricity board. Neither the project proponent nor the site personnel have any control over it. The records will be crosschecked with the records of sold electricity to state electricity board. The meters are calibrated by state electricity board at-least once in five years.

<u>Apportioning</u>

In case the start and end dates of a particular monitoring period do not match with the dates of the billing period, the net electricity exported to the grid would be calculated from the difference of number of days which are not matching of billing period and monitoring period or apportioned in line to the daily generation values for the said mismatch period.

The calculated value after apportioning would be used for calculation of emission reductions during that period.

Line Diagram



# APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Not Applicable as there is no commercially sensitive information is included in the project description.

# APPENDIX 2: CALIBRATION DETAIL

| Sl. No. | Purpose | Energy Meter | | | | Initial Meter Testing Date | Next Calibration Date |
|---|---|---|---|---|---|---|---|
| | | Serial Number | Make | Accuracy Class | Model | | |
| 1 | Main Meters | Q0944018 | SECURE | 0.2S | Apex 100 | 31-January-2024 | 30-January-2025 |
| 2 | | Q0944019 | | | | | |
| 3 | | Q0944020 | | | | | |
| 4 | | Q0944021 | | | | | |
| 5 | Check Meters | Q0944022 | | | | | |
| 6 | | Q0944023 | | | | | |
| 7 | | Q0944024 | | | | | |
| 8 | | Q0944025 | | | | | |